

Logical Observation Identifiers Names and Codes (LOINC®)



LOINC Mapping Guide: Allergy

VERSION 1.0

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loinc.org

Download the free LOINC database, access learning resources, ask questions, request new LOINC terms, and participate in the LOINC community.

search.loinc.org

Perform searches on the LOINC database using advanced syntax capabilities.

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Introduction

Logical Observation Identifiers Names and Codes (LOINC[®]) is a freely available international standard for identifying health measurements, observations, and documents. LOINC facilitates the storage, exchange, and pooling of results for clinical care, outcomes management, and research by providing a set of standard names and identifiers that can be used across heterogeneous computing environments.

At present, LOINC is used in more than 170 countries, of which over 30, including the United States, have adopted LOINC as a national standard. In the U.S. LOINC is specified for several key areas, including laboratory, in the 2019 U.S. Core Data for Interoperability. And in June 2018, the U.S. Food and Drug Administration (FDA) published Guidance that supports the adoption of LOINC for in vitro diagnostic (IVD) tests, the distribution of LOINC codes by IVD vendors, and the inclusion of LOINC codes for IVD tests in product labeling.

This guide is one in a series of guides intended to help users choose the correct LOINC code in a particular laboratory domain. The final published version will be available in the [Knowledge Base](#) section of the LOINC website.

Purpose

This document provides examples and guidance for selecting LOINC terms for laboratory tests related to Allergy. Specifically, it will cover mapping panel orders and observation values reported to a patient record. The audience includes vendors of *in vitro* diagnostic (IVD) assays, laboratory professionals and others tasked with mapping laboratory catalogs to LOINC, and anyone needing guidance on choosing the appropriate LOINC term for an individual assay within the field of Allergy.

Scope

This guide covers the general principles of choosing the correct laboratory LOINC term and mapping observation result values regardless of laboratory domain, as well as those specific to Allergy. It does not discuss determining order LOINC terms but may include order LOINC terms to help identify result mappings.

Before You Begin

Health Level Seven (HL7[®]) messaging examples demonstrate how LOINC-encoded results can be exchanged amongst information systems, applications, etc. At the time this guide was written, HL7 version 2 (HL7 v2), is widely used within the laboratory medicine community for messaging lab order and result information to and from laboratories and their laboratory information systems (LIS). HL7 Clinical Document Architecture (CDA[®]) messaging specifications are commonly used to communicate health information from Electronic Health Record (EHR) systems, including lab orders and results and their respective LOINC mappings.

HL7 Fast Healthcare Interoperability Resources (FHIR®) standards adoption has quickly spread globally for exchanging health data, especially with applications within health information systems and those external to traditional healthcare infrastructure such as insurance, consumer, and patient facing applications. Readers are advised to refer to the most recent HL7 Implementation Guides for complete specifications of how laboratory data and their respective LOINC codes are communicated with HL7 standards. Given its widespread use in laboratory result reporting, the focus and examples in this guide are on HL7 v2.

LOINC continues to evolve following changes in laboratory medicine as well as those regarding best practices for reporting, storing, and aggregating laboratory data. In some cases, LOINC terms that represented the preferred way of reporting 15 or 20 years ago may not necessarily be the best concepts to use today based on current understanding. In such cases, we still present information about mapping to these concepts, but with a usage note titled “Evolving model” (to indicate describing the mapping strategy that is no longer preferred) along with recommendations for which codes to use based on current best practices. Keep in mind some older LOINC codes may be mapped to historic laboratory data in information systems. We anticipate publishing new versions of this guide with new LOINC content and updates to mapping best practices as they develop. Comments, proposed edits, or other suggestions for future versions of this guide are welcome through our [dedicated feedback site](#). For questions or help unrelated to this guide, please contact us via the [Contact LOINC](#) page.

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Part 1: General Principles for Mapping to Laboratory LOINC

1 Before Beginning the Mapping Process

1.1 Utilize Existing LOINC Mappings

Before starting a new mapping effort, investigate whether any resources exist that may be of some help. Some tests in the compendium may already have LOINC codes supplied by the instrument or assay vendor, and major reference laboratories' LOINC mappings can be used to help inform the local mapping for similar tests. It is a good idea to perform an audit of the supplied LOINC codes against the existing assays' attributes before mapping your own.

1.1.1 In Vitro Diagnostics (IVD) Vendors

Many IVD vendors have mapped, or are in the process of mapping, their orders and test results to LOINC terms. Some of these mappings are publicly available, but more commonly the specific maps for a particular device are reserved for customers who purchase that device. Before you begin mapping or updating your maps, speak to your IVD representative to find out if LOINC mappings are available. The list provided by the vendor will likely cover all approved uses of the test kit, not just those implemented in your laboratory.

As more and more IVD vendors begin to adopt the LOINC for In Vitro Diagnostics (LIVD) format as a standard, integrating LOINC mappings from multiple vendors will become more streamlined.

1.1.2 Reference Laboratories

In the United States, many large reference laboratories, including Mayo Medical Laboratories, ARUP Laboratories, LabCorp, and Quest Diagnostics, publish LOINC mappings for each test in their online catalog. They can typically be found as a downloadable file or on the individual page for each lab test. Reference laboratory mappings are a great place to start your initial mapping or to use as a quality check for existing mappings. Keep in mind, however, that different laboratories may not perform the same test or report results in the same way. Be sure to review the test details provided by the performing laboratory to ensure it matches the test you are mapping. If a laboratory does not provide a LOINC code or map, ask their representatives if LOINC codes are available.

1.1.3 National Health Systems and Ministries of Health

In some countries, including Austria, Belgium, Canada, France, Portugal, the Netherlands, and others, resources are available at a national level. Such resources include databases with constrained subsets of LOINC terms applicable within the context of that country and lists of tests and their associated LOINC terms. Several countries also publish mappings from their national billing codes to LOINC. Before you begin the mapping process, make sure you understand the requirements related to LOINC usage in each country or jurisdiction as these may significantly affect your mapping strategy.

LOINC content and documentation have been [translated](#) into many languages, courtesy of volunteer translators, all of which may be helpful resources with country-specific usages of LOINC.

1.2 Mapping Tools

Two tools are available to aid in searches for LOINC codes. They are RELMA and search.loinc.org. Each has features that may be beneficial depending on the needs of the user.

1.2.1 Regenstrief Electronic LOINC Mapping Assistant (RELMA)

[RELMA](#) is a desktop application created specifically for LOINC mapping and is maintained by the LOINC software development team. RELMA's main purpose is to assist users with mapping local test codes (mnemonics) to LOINC codes. It is a powerful tool with features to import your catalog, data dictionary, or file of tests and map each to a specific LOINC term. It stores these mappings and provides multiple export options for loading back into your laboratory information system (LIS), electronic specimen collection manual, or other similar system. RELMA is highly recommended for a single person mapping more than a few terms.

It does have some limitations. RELMA is based on older technology and only runs on the Microsoft Windows operating system for an individual user. Mac or Linux users must configure a virtual machine with Windows to use it. (Instructions for this are [available on the LOINC website](#).) New versions of RELMA are released twice a year, in conjunction with LOINC database releases each August and February, and must be installed after each release to stay current. Given these limitations, the LOINC team is considering the development of alternatives to the current RELMA format.

1.2.2 Online Search Application

Our online search application (<https://search.loinc.org>) is fast, easy to use, and runs from an internet browser with no installation necessary. It does not have all the mapping features of RELMA but is useful for quick LOINC searches. It is also updated twice a year with each LOINC release and only requires a LOINC user login to access.

1.3 LOINC Formatting Notes

1.3.1 Part Types and Names

The types of LOINC *Parts*, such as *Component* and *Property*, are capitalized and italicized. The names of LOINC *Parts* that appear in LOINC term names, such as **Urine** and **Ord**, are capitalized and set in a bold typeface. For more information regarding the relationship between LOINC terms and LOINC *Parts*, please refer to [Section 2](#) Major Parts of a LOINC term section in the LOINC Users' Guide.

1.3.2 LOINC Format

The LOINC Fully Specified Name (FSN) is composed of the six primary LOINC *Parts* concatenated with colons (":") as delimiters. The Long Common Name (LCN) is an algorithmically created name that is unique for each LOINC term. When specific LOINC terms are referenced, they are presented as the LOINC number and FSN followed by the LCN. For example:

[54347-0](#) Albumin:SCnc:Pt:Ser/Plas:Qn:
LCN: Albumin [Moles/Vol] in Serum

1.3.3 Tall Man Casing

LOINC *Components* are not case sensitive. However, some LOINC terms, typically those containing drug names, may use tall man lettering. This is based on recommendations from the Institute for Safe Medication Practices. The casing does not imply any meaning; it simply draws attention to the unique characters in drugs with similar names. Some examples would be cycloSPORINE, cycloSERINE, chlorproPAMIDE induced platelet Ab.IgG, or chlorproMAZINE induced platelet Ab.IgG. Regardless of the casing used, the meaning should remain the same to prevent confusion or miscommunication in data exchange.

2 Choosing the Right LOINC Term (Mapping)

Although choosing the correct LOINC term for a given chemistry test can seem daunting, with a systematic approach and use of available resources, mapping can be straightforward for most concepts. Keep in mind, mapping from the perspective of an end user is different from mapping as an IVD vendor. A vendor should map to all the LOINC codes that support the range of specimens and reporting options approved for a single test kit, while end users should map to the set of LOINC codes that are appropriate for a particular implementation. For additional information about mapping from a vendor perspective and examples of the differences between vendor and end-user mappings, refer to the domain-specific section of this guide.

2.1 General Mapping Principles

The three most important overarching principles are:

1. Map to the most specific LOINC term possible based on the available information.
2. Do not overspecify by assuming information that is not known.
3. If a LOINC does not exist for your test information, request a new LOINC as soon as possible.

For example, if you are mapping a test named Benzthiazide random urine, a search of just “Benzthiazide” will generate a list of the following candidate terms:

LOINC Benzthiazide Search												
RESULTS	DISPLAYING	FILTER	VIEW	EXPORT								
6	1-6		List Card									
Status	LOINC	Long Common Name	Component	Property	Timing	System	Scale	Method	Class	Type	Example UCUM Units	Order/Observation
	4141-8	Benzthiazide [Mass] of Dose	Benzthiazide	Mass	Pt	Dose	Qn		DRUGDOSE		mg	Observation
	3399-3	Benzthiazide [Presence] in Urine	Benzthiazide	PrThr	Pt	Urine	Ord		DRUG/TOX			Both
	3400-9	Benzthiazide [Mass/volume] in Urine	Benzthiazide	MCnc	Pt	Urine	Qn		DRUG/TOX		ug/mL	Both
	43863-0	Benzthiazide [Presence] in 24 hour Urine	Benzthiazide	PrThr	24H	Urine	Ord		DRUG/TOX			Both
	3401-7	Benzthiazide [Mass/time] in 24 hour Urine	Benzthiazide	MRat	24H	Urine	Qn		DRUG/TOX		ug/(24.h)	Both
	16449-1	Benzthiazide [Mass/volume] in Serum or Plasma	Benzthiazide	MCnc	Pt	Ser/Plas	Qn		DRUG/TOX		ug/mL	Both

However, if you search for “Benzthiazide Timing:24H”, you will see a shorter list of candidate terms.

LOINC		Benzthiazide										Search			
RESULTS		DISPLAYING		200		FILTER		VIEW		EXPORT					
6		1-6				T		List Card		B					
Status															
LOINC	Long Common Name					Component	Property	Timing	System	Scale	Method	Class	Type	Example UCUM Units	Order/Observation
4141-8	Benzthiazide [Mass] of Dose					Benzthiazide	Mass	Pt	Dose	Qn		DRUGDOSE	mg		Observation
3399-3	Benzthiazide [Presence] in Urine					Benzthiazide	PrThr	Pt	Urine	Ord		DRUG/TOX	mg		Both
3400-9	Benzthiazide [Mass/volume] in Urine					Benzthiazide	MCnc	Pt	Urine	Qn		DRUG/TOX	ug/mL		Both
43863-0	Benzthiazide [Presence] in 24 hour Urine					Benzthiazide	PrThr	24H	Urine	Ord		DRUG/TOX	mg		Both
3401-7	Benzthiazide [Mass/time] in 24 hour Urine					Benzthiazide	MRat	24H	Urine	Qn		DRUG/TOX	ug/(24.h)		Both
16449-1	Benzthiazide [Mass/volume] in Serum or Plasma					Benzthiazide	MCnc	Pt	Ser/Plas	Qn		DRUG/TOX	ug/mL		Both

Which LOINC term is most appropriate? Notice that your test name specifies 24H timing. With this information, you can feel confident that [43863-0](#) *Benzthiazide [Presence]* or [3401-7](#) *Benzthiazide [Mass/time]* is the best LOINC for your test, depending on the test property desired.

As another example, if you are mapping a test named “Benzthiazide” system of urine and you search for “Benzthiazide System:Urine”, you will see a short list of candidate terms specific to the criteria.

LOINC ▼ Benzthiazide System:Urine Search												
RESULTS		DISPLAYING	FILTER	VIEW	EXPORT							
4		200	1-4	Y	List Card							
Status	LOINC	Long Common Name	Component	Property	Timing	System	Scale	Method	Class	Type	Example UCUM Units	Order/Observation
	3399-3	Benzthiazide [Presence] in Urine	Benzthiazide	PrThr	Pt	Urine	Ord		DRUG/TOX			Both
	3400-9	Benzthiazide [Mass/volume] in Urine	Benzthiazide	MCnc	Pt	Urine	Qn		DRUG/TOX		ug/mL	Both
	43863-0	Benzthiazide [Presence] in 24 hour Urine	Benzthiazide	PrThr	24H	Urine	Ord		DRUG/TOX			Both
	3401-7	Benzthiazide [Mass/time] in 24 hour Urine	Benzthiazide	MRat	24H	Urine	Qn		DRUG/TOX		ug/(24.h)	Both

Suppose you are mapping a test named Beta 1 globulin Urine Mass Concentration (**Mcnc**) Electrophoresis and you search for “Beta 1 globulin Urine,” you will see the following list of candidate terms.

LOINC ▼ Beta 1 globulin Urine Search												
RESULTS		DISPLAYING	FILTER	VIEW	EXPORT							
17		200	1-17	Y	List Card							
Status	LOINC	Long Common Name	Component	Property	Timing	System	Scale	Method	Class	Type	Example UCUM Units	Order/Observation
	99600-9	Beta 1 globulin [Mass/volume] in 24 hour Urine by Electrophoresis	Beta 1 globulin	MCnc	24H	Urine	Qn	Electrophoresis	CHEM		g/L	Both
	32734-6	Beta 1 globulin [Mass/time] in 24 hour Urine by Electrophoresis	Beta 1 globulin	MRat	24H	Urine	Qn	Electrophoresis	CHEM		g/(24.h)	Observation
	9829-3	Beta globulin [Mass/time] in 24 hour Urine	Beta globulin	MRat	24H	Urine	Qn		CHEM		g/(24.h)	Observation
	54353-8	Beta 1 globulin [Mass/volume] in Urine by Electrophoresis	Beta 1 globulin	MCnc	Pt	Urine	Qn	Electrophoresis	CHEM		g/L	Both
	77342-4	Beta 1 globulin [Mass/volume] by Electrophoresis in Urine collected for unspecified duration	Beta 1 globulin	MCnc	XXX	Urine	Qn	Electrophoresis	CHEM		mg/dL	Observation
	6889-0	Beta globulin [Mass/volume] in 24 hour Urine	Beta globulin	MCnc	24H	Urine	Qn		CHEM		g/dL	Observation
	29951-1	Beta globulin [Presence] in 24 hour Urine	Beta globulin	PrThr	24H	Urine	Ord		CHEM			Observation

Notice that your test name specifies mass concentration for the *Property* as well as Electrophoresis as the *Method*. With those two pieces of information, you can feel confident that 54353-8 Beta globulin 1 [Mass/volume] in Urine by Electrophoresis is the best LOINC for your test.

Equally important, however, is not to overspecify by guessing about information you do not have. As seen in the previous example, LOINC has many similar terms for the same globulin that vary by subtle distinctions. These distinctions are important because they reflect the differences in clinical utility for each test (e.g., the collection of 24-hour urine samples vs non-challenge). Mapping accurately requires attention to these details and not making assumptions about the test being mapped.

As we see below, searching for the same test but adding electrophoresis to your search term will lessen the list of LOINC candidates returned.

LOINC ▼ Beta 1 globulin Urine electrophoresis Search												
RESULTS		DISPLAYING	FILTER	VIEW	EXPORT							
13		200	1-13	Y	List Card							
Status	LOINC	Long Common Name	Component	Property	Timing	System	Scale	Method	Class	Type	Example UCUM Units	Order/Observation
	99600-9	Beta 1 globulin [Mass/volume] in 24 hour Urine by Electrophoresis	Beta 1 globulin	MCnc	24H	Urine	Qn	Electrophoresis	CHEM		g/L	Both
	32734-6	Beta 1 globulin [Mass/time] in 24 hour Urine by Electrophoresis	Beta 1 globulin	MRat	24H	Urine	Qn	Electrophoresis	CHEM		g/(24.h)	Observation
	54353-8	Beta 1 globulin [Mass/volume] in Urine by Electrophoresis	Beta 1 globulin	MCnc	Pt	Urine	Qn	Electrophoresis	CHEM		g/L	Both
	77342-4	Beta 1 globulin [Mass/volume] by Electrophoresis in Urine collected for unspecified duration	Beta 1 globulin	MCnc	XXX	Urine	Qn	Electrophoresis	CHEM		mg/dL	Observation
	97764-5	Beta 1 globulin/Protein.total in 24 hour Urine by Electrophoresis	Beta 1 globulin/Protein.total	MFr	24H	Urine	Qn	Electrophoresis	CHEM		%	Both
	38186-3	Beta globulin [Mass/volume] in 24 hour Urine by Electrophoresis	Beta globulin	MCnc	24H	Urine	Qn	Electrophoresis	CHEM		g/dL	Observation

Based on the results from the search Beta 1 globulin urine electrophoresis, we can eliminate those with a timing component specified as 24H because the timing was not requested in the search. We are now left with two tests. [77342-4](#) is not the right one because duration is unspecified. The right choice will be mapping to [54353-8](#) *Beta 1 globulin [Mass/volume] in Urine by Electrophoresis*.

KEY POINTS

- Use all available information.
- Do not make assumptions about test characteristic information you do not have.

2.2 Minimum Test Information Necessary to Map

Care must be taken to understand the contexts, details, specimen information, and laboratory methods underlying tests being mapped. Begin by assembling information associated with how the test is performed. If you are not the performing laboratory, you may need to ask the performing laboratory for their LOINC codes or to clarify how the test is performed, if this information is not readily available on their website or in their CLIA Specimen Collection Manual. These documents typically include the specific analyte being measured, the units of measure, how the results are reported, what specimens can be used to measure the analyte, and other information needed to map to a specific LOINC.

The subsections which follow highlight information commonly needed for each lab test. Assemble any reference laboratory LOINC maps, LIVED LOINC maps for instruments or test kits used in testing, IVD manufacturer package inserts, and any information on how your information system supports LOINC mapping. It is often helpful to compare similar tests and determine the appropriate mapping for each.

While it is not necessary to map all tests at the same time, it is usually more efficient, especially when mapping a large number of tests that follow the same pattern. Your Information Technology team can help with an export of tests from your information system (if loading into RELMA for mapping), as well as any specifications and requirements to upload mapped data back into your information system to help minimize rework later.

If you are using a terminology vendor product, you'll want to have their resources at hand as you look up LOINC codes in their products or tooling to aid for quicker matches and their mapping process. You may also wish to have product support guides and any vendor references to assist in the process.

Package inserts and Standard Operating Procedures (SOP) for commercially- and laboratory-developed tests, respectively, provide invaluable information for mapping tests to LOINC codes. The CLIA specimen collection manual, laboratory test compendium, or online directory of service also includes this information.

2.2.1 Analyte, Specimen, and Method

Key information necessary for choosing the correct LOINC term includes the name of the analyte (*Component*), the specimen type (*System*), and if relevant, the *Method*.

This information is usually found in the package insert or SOP, or if working primarily from a laboratory catalog or LIS system, many of these details are in the test name. The test *Method* may only be specified for certain terms in which the particular method is significant for ordering the test and/or interpreting the result. If working with test names that do not include analyte or specimen type, it is vital to locate that information before you begin. In cases, where a test result can be performed on different specimen types, specimen or source information may be indicated in another field. The allowed specimen(s) to be collected and accepted by the performing laboratory should be in the CLIA specimen collection manual.

2.2.2 Example Results

Example results and associated information, such as average values and reference ranges, are extremely useful for differentiating tests that seem identical or similar based on test name alone. For example, they can help distinguish between tests that have qualitative results that are ranked, such as 1+, 2+, 3+ (LOINC *Scale Ord*), and those that cannot be ranked, such as type of Adenovirus (LOINC *Scale Nom*). If example test reports are unavailable, the result type or data type indicating whether the expected results are quantitative based on an answer list, free text, etc. can be useful information. A single assay may be approved for multiple reporting options. These will be described in the package insert. Some assays may produce a quantitative result value, but it is reported as a qualitative interpretation (e.g., positive or negative). It is therefore important to understand how the result is reported so accurate LOINC mapping can occur.

2.2.3 Units of Measure

Units of measure are critical for mapping to the correct LOINC term. For quantitative results, LOINC has different terms for results reported in mass units versus molar units such as mass concentration (mg/dL) and substance (molar) concentration (mol/L). Without units of measure, you may not know which quantitative LOINC term is appropriate.

Some tests may be reported with a single unit of measure across a country while other tests may use different units within the same country. Some laboratories may also report result values in the same units of measure, but using difference scales (e.g., mg/dL versus mg/L). They are often mapped to the same LOINC as they are both mass concentration, but if the result values are used in calculations, they may need to be converted to the same scale to be usable. Some IVD vendors will convert their result values to different scales on their instruments depending on how a laboratory has the instrument configured. In other cases, conversions may occur in the LIS or another information system.

KEY POINTS

- The Package Inserts and Standard Operating Procedures are excellent sources of information.
- The following data are essential for choosing the correct LOINC term:
 - *Analyte*
 - Specimen(s) analyzed
 - Qualitative sample results
 - Units of measure for quantitative results
 - *Method* of analysis, if relevant

2.3 Additional Useful Data

2.3.1 List of Abbreviations Used by Your Laboratory in Test Names

Each organization tends to use its own conventions for abbreviations in local test names or mnemonics. Many abbreviations are self-explanatory, but inevitably there are a few ambiguous ones. Having a list of those abbreviations will save time and frustration during the mapping process. Examples of abbreviations that may cause confusion amongst different laboratories are highlighted by Hauser et al¹. It is highly recommended to use the Unified Code for Units of Measure (UCUM) to represent units in a standardized format. Further details about UCUM are in the [Properties and Units of Measure](#) and [Related standards](#) sections.

2.3.2 Frequency Data

Test usage frequency may be helpful when mapping in some cases and not as helpful in others. A small number of tests usually represent a relatively high percentage of the test volume in a typical lab, and such high-volume tests should probably be given higher mapping priority. Test results important for public health and safety and/or those required to be reported under local regulations should also be prioritized. Business and customer needs may also dictate prioritization of mapping some tests before others. With increasing requirements spurring the adoption of LOINC, many laboratories focus on priority items first, then work to encode most if not all their remaining test results and orders.

For new assays coming to market, it is recommended that the IVD manufacturer assess whether an existing LOINC code can be mapped to the assay, or if a request for a new LOINC code is needed. When a new specimen, method (such as with consumer-based testing), or result report occurs, a new LOINC may be warranted. Given the time needed to create a new LOINC and the frequency of releases, requests should be made as soon as possible.

1 Ronald George Hauser, M. D.-A. (2022, June 17). Development and Implementation of a Standard Format for Clinical Laboratory Test Results. Retrieved from Oxford Academic: <https://academic.oup.com/ajcp/article/158/3/409/6609855>

2.4 Part-specific Mapping

The six *Parts* that make up a LOINC term are *Component*, *Property*, *Timing*, *System*, *Scale*, and *Method*.

2.4.1 Component

The *Component* is the most important axis for mapping because it describes the specific analyte being measured. A given test only measures one analyte and it is important to choose the LOINC term with the *Component* that reflects what is being measured. Use caution when looking at the *Component* as terms with similar *Components* could potentially represent very different concepts. For example, the *Components* **Erythropoietin** and **Erythropoietin Ab** look quite similar, however the first represents the glycoprotein and the second represents immunoglobulins to the glycoprotein. These two concepts are related in that exposure to the glycoprotein could lead to an immune response and production of antibodies against the glycoprotein.

As another example, the *Components* **Methylenedioxyethylamphetamine** and **Methylenedioxymethamphetamine** also look very similar. However, the first represents the drug and the second represents antibodies to the drug. These two concepts are related in that exposure to the drug could lead to an immune response and production of antibodies against the drug. However, most tests only measure a single analyte, and it is important to choose the LOINC term with the *Component* that reflects what is being measured. Blood banking terms can be similarly confusing as there are a parallel set of *Components* that represent red blood cell antigens and antibodies to those antigens, such as **B Ag** and **B Ab**. Map to what is being tested for, and not by the reagent name. The B antisera reagent is testing for the B antigen on the patient's cells. Be careful when mapping to such terms.

2.4.1.1 Structure of LOINC Components

A single LOINC *Component* may have multiple subparts. What you see in the *Component* field in RELMA or the online search application is the primary *Component*, sometimes referred to as the “full” *Component*. In some cases, there are no subparts, and the primary *Component* is the same as the core *Component*. However, in other cases, the primary *Component* consists of a core *Component* plus additional subparts, such as a *Challenge*, *Adjustment*, or *Divisor*. The same core *Component* can be used with a variety of additional subparts to create unique analytes.

The following table illustrates the structure of a variety of laboratory *Components*. Please see [Section 2.2](#) (*Component/Analyte*) of the LOINC User's Guide for further discussion of these structures.

Primary Component	Core Component	Challenge	Adjustment	Divisor
Ketones	<i>Ketones</i>	-	-	-
Ketones^1H post 50 g glucose PO (measured 1 hour after a 50-gram oral glucose challenge)	<i>Ketones</i>	<i>1H post 50 g glucose PO</i>	-	-
Calcium.ionized^adjusted to pH 7.4 (ionized calcium result is adjusted for a pH of 7.4)	<i>Calcium.ionized</i>	-	<i>adjusted to pH 7.4</i>	-
Glutamate/Amino acids.total (fraction of glutamate out of total amino acids)	<i>Glutamate</i>	-	-	<i>Amino acids.total</i>
Glucose^1H post meal (glucose 1 hour after a meal)	<i>Glucose</i>	<i>1H post meal</i>	-	-
cycloSPORINE	<i>cycloSPORINE</i>	-	-	-
cycloSPORINE^4H post dose	<i>cycloSPORINE</i>	<i>4H post dose</i>	-	-
7-Hydroxyquetiapine^^adjusted for lean body mass+urine creatinine	<i>7-Hydroxyquetiapine</i>	-	<i>adjusted for lean body mass+urine creatinine</i>	-
2-Hexanol/Creatinine	<i>2-Hexanol</i>	-	-	<i>Creatinine</i>
Valproate.free/Valproate.total	<i>Valproate.free</i>	-	-	<i>Valproate.total</i>

2.4.2 Challenge Test

The second subpart contains information necessary to interpret “challenge” (or loading or tolerance) tests. Variables that report the result of a measurement taken a certain amount of time post-challenge (e.g., glucose after an oral glucose tolerance test) must be distinguished according to the challenge and the time post-challenge. In general, it is recommended to use separate observation terms for results that are measured separately.

The *Challenge* subpart has a substructure that identifies the time interval for, or time difference from, the challenge. It uses the following syntax, where the word “post” (or base line) is required: <time delay> “post” <challenge>. The challenge can be further characterized as <amount given> <substance/treatment given> <route given>. An example of a *Challenge* that used all parts would be “Aldosterone^1H post 25 mg captopril PO.”

The time difference follows the syntax $n<S|M|H|D|W>$, where n is a number (possibly a decimal) and S denotes seconds, M denotes minutes, H denotes hours, D denotes days, and W denotes weeks. The time delay can be preceded by a greater than ($>$) sign (e.g., $>4H$). Any time specification that follows this syntax is permitted.

In addition to specifying an amount of time elapsed since the challenge, the time delay slot can be used to name an actual time when the measurement was taken (e.g., Glucose^{10:00 AM} specimen) or to specify the ordering of specimens (e.g., ^{1st} specimen, ^{2nd} specimen). Use this syntax to indicate pre- and post-immunization specimens, acute and convalescent specimens, or a series of specimens for which no more detailed information is available.

The second subpart is also used to describe measurements taken at a specified point after the start of an ongoing treatment, such as peritoneal dialysis (e.g., Creatinine^{12H} post peritoneal dialysis). More generally, this syntax can be used to indicate that observations were recorded (e.g., ^{post-partum}, ^{postoperative}, or ^{post EDTA therapy}).

The syntax of the second subpart can be specified in various ways to indicate *Challenges* of greater or lesser specificity, based on the amount of detail that the laboratory knows about the challenge specimen.

2.4.2.1 Generic *Challenge* Specifications

Specificity of *Challenges* can range from fully specified (e.g., ^{30M} post 100 g glucose PO) to generic (e.g., ^{post XXX} challenge). Partial *Challenges* include the following:

- no amount specified (e.g., ^{30M} post dose glucose)
- a time elapsed but not a particular challenge (e.g., ^{1H} post XXX challenge)
- no exact time but with ordering information (e.g., ^{2nd} specimen post XXX challenge) or even more generic (^{post XXX} challenge)

These latter variants are needed to accommodate challenges that do not fit any common protocol or refer to reference laboratories where the study protocol is not reported.

2.4.2.2 Operators

Specific operators are used to differentiate certain *Component* subparts.

- Slash (/): separates numerator from denominator, for example, Beta-N-acetylhexosaminidase.B/Beta-n-acetylhexosaminidase.total in Blood. (Note that *Components* with a slash are usually associated with a fraction or ratio *Property*.)
- Carat (^): separates the core *Component* from:
 - *Challenge* (after the first carat)
 - Adjustment (after the second carat – rarely used)
 - Count (after the third carat – even more rare)
- Dot (.): permits a subclass or further detail about an analyte (e.g., Beta-N-acetylhexosaminidase.B indicates the “B” isozyme of Beta-N-acetylhexosaminidase)
 - Note: The dot is part of the core *Component* structure. It does not separate different subparts.

2.4.2.3 Ampersand vs. Plus

Be careful when mapping to *Components* that contain plus (+) or ampersand (&). In LOINC, the “+” indicates that the test does not differentiate between the analytes when reporting the result. The “&” indicates that it does differentiate.

KEY POINTS

- Ampersand (&) indicates that multiple analytes can be differentiated.
- Plus (+) indicates that multiple analytes cannot be differentiated.

For example, HIV 1+2 Ab indicates a test detects antibodies to both HIV 1 and HIV 2 but cannot differentiate between them when reporting the result. A result value of “Detected” means that antibodies against HIV 1 and/or HIV 2 were detected.

In contrast, Chlamydia trachomatis & Neisseria gonorrhoeae DNA indicates a test that detects both Chlamydia trachomatis and Neisseria gonorrhoeae but can differentiate between the two. In this test result values might include “Chlamydia trachomatis not detected; Neisseria gonorrhoeae detected” or “Neither Chlamydia trachomatis nor Neisseria gonorrhoeae detected.”

EVOLVING MODEL

Historically, tests that produced two or three individual results that were reported together in one field were modeled in LOINC by a single term, where all the analytes were named and separated – by “&” in the *Component* and “and” in the Long Common Name. This model is no longer recommended because the individual analyte results are not computable. Now, only assays that truly produce a single result are modeled this way; otherwise, a panel will be created with individual children for each result.

2.4.3 Scale and Property

The LOINC axes *Scale* and *Property* are closely related. *Scale*, previously called precision, represents a higher-level categorization of how a result is reported, such as quantitative or qualitative, where quantitative (**Qn**) is numeric and qualitative is either ordinal (**Ord**) or nominal (**Nom**). *Scale* abbreviations and their definitions can be found in Table 12, [section 2.6](#) Type of *Scale* (5th part) of the LOINC Users Guide.

Since *Scale* represents a broad category, it can be used to quickly narrow down the list of candidate LOINC terms. For example, if you know the test result is qualitative, you can immediately eliminate all the LOINC terms that represent quantitative measurements. Once you have a more manageable set of terms, you can focus on the specific distinctions of the *Property*.

Property, also called kind of quantity, describes the specific aspect of the analyte that is being measured, such as its mass concentration or volume. A list of example LOINC *Properties* can be found in section [2.3.3, Table 8](#) of the LOINC Users Guide.

The following table illustrates how *Property* and *Scale* differ for various types of results. This is a small sample of the many *Property* types and *Scale* combinations. A larger list of *Property*

abbreviation examples can be found in [Appendix E](#) of the LOINC Users Guide. The complete set of active LOINC *Properties* is available in the LOINC Part File (identified with the PartTypeName of PROPERTY).

Type of result value	LOINC Property	LOINC Scale	LOINC term examples
Presence/absence, OR graded results	PrThr	Ord	8030-9 Streptococcus pneumoniae Ab: PrThr:Pt:Ser:Ord <i>Example: Present, Absent</i>
Absence or, if present, the specific type or identity of what is present	Prid	Nom	34601-5 Drugs identified:Prid:Pt:Body fld:Nom <i>Example: None, Amphetamine, Barbiturate, Oxycodone</i>
Identification of the type of analyte that is already known to be present	Type	Nom	882-1 ABO & Rh group:Type:Pt:Bld:Nom <i>Example results: O pos, O neg, A pos, A neg</i>
Quantitative	ACnc	Qn	80221-5 N-methyl-D-aspartate receptor Ab.IgG: Titr:Pt:Ser:Qn:IF <i>Example: 1:4</i>
Impression/ Interpretation	Imp	Nom	88700-0 Babesia microti Ab.IgG & IgM:Imp:Pt:Ser:Nom: <i>Example: Ab not detected, Recent infection, Past infection</i>

2.4.3.1 Quantitative Properties

Quantitative *Properties* describe measurable aspects of an analyte (e.g., mass, counts, concentrations, etc.) and are always associated with a *Scale* of **Qn**.

Quantitative Property	Abbreviation
Mass Concentration	MCnc
Substance Ratio	SRto
Mass Rate	MRat
Volume Fraction	VFr

2.4.3.1.1 Fraction vs. Ratio

Fraction and ratio are two quantitative *Properties* in LOINC that are often confused by users. Although both typically have a numerator and denominator, there is a significant difference. Fractions represent a part of the whole. An analyte in the numerator represents a portion, or subset, of the analyte in the denominator. In contrast, ratios represent the comparison of two quantities where one is not a subset of the other.

The following are a few examples of fraction and ratio *Property* type *Components*. Note that the specific *Property* depends on the aspect of the analytes being measured (mass, substance, volume, etc.).

<i>Component</i>	<i>Property type</i>
Glutamate/Amino acids.total	Fraction
Glutamate/Creatinine	Ratio
Albumin/Protein.total	Fraction
Albumin/Globulin	Ratio
Coproporphyrin 3/Porphyrins.total	Fraction
IgA.kappa/IgA.lambda	Ratio

2.4.3.1.2 Rate vs. Ratio

Similarly, users often confuse LOINC rate and ratio *Properties* because they look alike. Rate *Properties* end with “Rat” and ratio *Properties* end with “Rto.” Be careful when mapping to these terms, paying specific attention to the units. Another way to distinguish between these is that the *Component* for terms representing ratios will often be displayed with a numerator and denominator, such as **Glutamine/Creatinine** or **Acylcarnitine/Carnitine.free**. Also the units of measure will include an analyte in the denominator (e.g., **nmol/g creatinine**). On the other hand, if the units-of-measure have a time period in the denominator, /min for example, this indicates a rate *Property*.

Suppose your lab offers a test called “24-hour urine Cortisol” and you have already determined that the analyte being measured is cortisol and not free cortisol. You search LOINC for that test name and find several candidate terms as shown below.

LOINC		24 hour urine cortisol -free		Search								
RESULTS	47	DISPLAYING	200	FILTER	VIEW							
		1-47			EXPORT							
					List Card							
Status	LOINC	Long Common Name	Component	Property	Timing	System	Scale	Method	Class	Type	Example UCUM Units	Order/Observation
	20622-7	Cortisol [Mass/volume] in 24 hour Urine	Cortisol	MCnc	24H	Urine	Qn		CHEM		mg/dL	Both
	14158-0	Cortisol [Mass/time] in 24 hour Urine	Cortisol	MRat	24H	Urine	Qn		CHEM		ug/(24.h)	Both
	50848-1	Cortisol [Moles/volume] in 24 hour Urine	Cortisol	SCnc	24H	Urine	Qn		CHEM		nmol/L	Both
	32310-5	Cortisol [Moles/time] in 24 hour Urine	Cortisol	SRat	24H	Urine	Qn		CHEM		nmol/(24.h)	Both
	95579-9	Cortisol [Z-score] in 24 hour Urine	Cortisol	Zscore	24H	Urine	Qn		CHEM		[Zscore]	Observation
	83089-3	Cortisol [Mass/volume] in 24 hour Urine by Immunoassay	Cortisol	MCnc	24H	Urine	Qn	IA	CHEM		ng/mL	Both
	83091-9	Cortisol [Moles/volume] in 24 hour Urine by Immunoassay	Cortisol	SCnc	24H	Urine	Qn	IA	CHEM		nmol/L	Both
	44310-1	Cortisol/Creatinine [Mass Ratio] in 24 hour Urine	Cortisol/Creatinine	MRto	24H	Urine	Qn		CHEM		ug/g[creat]	Both
	13611-9	6-Beta-Hydroxycortisol [Mass/time] in 24 hour Urine	6-Beta-Hydroxycortisol	MRat	24H	Urine	Qn		CHEM		ug/(24.h)	Both

Which term is the correct match? It cannot be determined from the information given. You do not know whether the result is reported in mass-based units or substance-based units, or whether it is a ratio, rate, or concentration. Finally, *Method* could be a factor if the test is an immunoassay.

You find out from your lab that this test is resulted in micrograms/24 hours which is mass over time giving it a *Property* of mass rate (**MRat**). Adding mass and rate or just **MRat** to the search will parry down the list significantly.

LOINC		24 hour urine cortisol -free mass rate		Search								
RESULTS		DISPLAYING	200	FILTER	VIEW	EXPORT						
13		1-13			List Card							
Status	LOINC	Long Common Name	Component	Property	Timing	System	Scale	Method	Class	Type	Example UCUM Units	Order/ Observation
	14158-0	Cortisol [Mass/time] in 24 hour Urine	Cortisol	MRat	24H	Urine	Qn		CHEM	👤	ug/(24.h)	Both
	13611-9	6-Beta-Hydroxycortisol [Mass/time] in 24 hour Urine	6-Beta-Hydroxycortisol	MRat	24H	Urine	Qn		CHEM	👤	ug/(24.h)	Both
	55906-2	Alpha cortolone [Mass/time] in 24 hour Urine	Alpha cortolone	MRat	24H	Urine	Qn		CHEM	👤	mg/(24.h)	Observation
	95589-8	Beta cortolone [Mass/time] in 24 hour Urine	Beta cortolone	MRat	24H	Urine	Qn		CHEM	👤	ug/(24.h)	Observation
	21044-3	5-Alpha tetrahydrocortisol [Mass/time] in 24 hour Urine	5-Alpha tetrahydrocortisol	MRat	24H	Urine	Qn		CHEM	👤	ug/(24.h)	Both
	57634-8	Alpha cortolone [Mass/time] in 24 hour Urine --pre dose dexamethasone	Alpha cortolone*pre dose dexamethasone	MRat	24H	Urine	Qn		CHAL	👤	mg/(24.h)	Observation
	57618-1	5-Alpha tetrahydrocortisol [Mass/time] in 24 hour Urine --pre dose	5-Alpha tetrahydrocortisol*pre dose dexamethasone	MRat	24H	Urine	Qn		CHAL	👤	mg/(24.h)	Observation

Now the correct code can be identified since only one test remains that looks at “Cortisol,” that being [14158-0](#). You may also gain reassurance since the code has a decent rank (#7558) among the top 20,000 LOINC terms used by other labs.

Now let’s say that the units were instead reported as micrograms per grams of creatinine. From this, you know that this has a *Property* of ratio in mass-based units. By including “mass” and “ratio”, or “**MRto**”, into the search, the list comes down to one item: [44310-1](#) *Cortisol/Creatinine [Mass Ratio] in 24-hour Urine* which would be the correct LOINC term for the test.

LOINC										24 hour urine cortisol -free mass ratio										Search																																																																																																													
RESULTS										1										DISPLAYING 200										FILTER										VIEW										EXPORT																																																																															
										1-1																				List										Card																																																																																									
Status										LOINC										Long Common Name										Component										Property										Timing										System										Scale										Method										Class										Type										Example UCUM Units										Order/Observation									
										44310-1										Cortisol/Creatinine [Mass Ratio] in 24 hour Urine										Cortisol/Creatinine										MRto										24H										Urine										Qn																				CHEM										🧪										ug/g[creat]										Both									

2.4.3.1.3 Property and Units of Measure

The LOINC *Property* can often be determined by the units of measure. The *Property* determines the kind of units that are reportable with that term. LOINC has different *Properties* for analytes measured in mass units versus substance units (e.g., milligrams versus millimoles), but does not have different *Properties* to represent different orders of magnitude, such as milligram versus gram. Thus, when mapping, users must choose a LOINC term with a *Property* that is consistent with the reporting units. The table below provides some examples of common laboratory units of measure and the corresponding LOINC *Property* and *Scale*.

Units of measure	Property	Description	Scale
mg/dL; ng/mL	MCnc	Mass concentration	Qn
mmol/L; nmol/dL	SCnc	Substance concentration	Qn
copies/mL; #/mL	NCnc	Number concentration	Qn
log copies/mL; log #/mL	LnCnc	Log number concentration	Qn
#/area; /[HPF]; /[LPF]	Naric	Number per area	Qn
mmol/mol{creatinine}	SRto	Substance ratio	Qn

%	NFr	Number fraction	Qn
mg/(24.h); mg/d; mg/h	MRat	Mass rate	Qn
nmol/g; mol/kg	SCnt	Substance content	Qn
IU/L; IU/mL; index value; EIA units; units/mL	ACnc	Arbitrary concentration	Qn

The **ACnc** (arbitrary concentration) *Property* represents the number of arbitrary units in a volume. Immunoassay results that are based on optical densities are commonly represented in LOINC with this *Property*. Because the units are truly arbitrary, they may be represented in a variety of ways by different laboratories and within LOINC, as shown in the table above.

Adding a *Property* to your search query is another easy way to further narrow the list of candidate LOINC terms. For example, if the test's units of measure are mg/dL, searching for LOINC terms with the *Property* **MCnc** (mass concentration) will significantly narrow the number of terms returned.

Including units in the search query is another easy way to further narrow the list of LOINC candidates. The search engine functionality returns terms with the *Property* corresponding to the units. More advanced users may wish to search by the *Property* directly. For example, if the test's units of measure are mg/dL, searching for LOINC terms with the *Property* **MCnc** will significantly narrow the number of terms returned.

To illustrate this further, consider the two searches shown here. The first shows the search query “Urine sodium” and the second shows “Urine sodium SCnc.” The second search returns less than a quarter of the candidate LOINC terms of the first. Although you must still choose between six candidate terms, the mapping is straightforward if you know the duration of the urine specimen collection.

LOINC Urine sodium Search												
RESULTS 34		DISPLAYING 200	FILTER	VIEW List Card	EXPORT 							
1-34												
Status	LOINC	Long Common Name	Component	Property	Timing	System	Scale	Method	Class	Type	Example UCUM Units	Order/Observation
	2955-3	Sodium [Moles/volume] in Urine	Sodium	SCnc	Pt	Urine	Qn		CHEM	mmol/L		Both
	57370-9	Sodium [Moles/volume] in 12 hour Urine	Sodium	SCnc	12H	Urine	Qn		CHEM	mmol/L		Both
	35678-2	Sodium [Moles/volume] in Urine collected for unspecified duration	Sodium	SCnc	XXX	Urine	Qn		CHEM	mmol/L		Both
	18377-2	Sodium [Moles/time] in 12 hour Urine	Sodium	SRat	12H	Urine	Qn		CHEM	mmol/(12.h)		Both
	58994-5	Sodium [Moles/time] in 18 hour Urine	Sodium	SRat	18H	Urine	Qn		CHEM	mmol/(18.h)		Both
	18376-4	Sodium [Moles/time] in 6 hour Urine	Sodium	SRat	6H	Urine	Qn		CHEM	mmol/(6.h)		Both
	32553-0	Sodium [Molar amount] in Urine collected for unspecified duration	Sodium	Sub	XXX	Urine	Qn		CHEM	mmol		Both
	21526-9	Sodium [Mass/volume] in 24 hour Urine	Sodium	MCnc	24H	Urine	Qn		CHEM	mg/dL		Both
	21527-7	Sodium [Mass/time] in 24 hour Urine	Sodium	MRat	24H	Urine	Qn		CHEM	mg/(24.h)		Both
	26763-3	Sodium [Mass/time] in 2 hour Urine	Sodium	MRat	2H	Urine	Qn		CHEM	mg/(2.h)		Both

LOINC Urine sodium SCnc Search												
RESULTS	7	DISPLAYING 200	FILTER	VIEW	EXPORT							
	1-7			List Card								
Status	LOINC	Long Common Name	Component	Property	Timing	System	Scale	Method	Class	Type	Example UCUM Units	Order/Observation
	2955-3	Sodium [Moles/volume] in Urine	Sodium	SCnc	Pt	Urine	Qn		CHEM	mmol/L		Both
	57370-9	Sodium [Moles/volume] in 12 hour Urine	Sodium	SCnc	12H	Urine	Qn		CHEM	mmol/L		Both
	35678-2	Sodium [Moles/volume] in Urine collected for unspecified duration	Sodium	SCnc	XXX	Urine	Qn		CHEM	mmol/L		Both
	21525-1	Sodium [Moles/volume] in 24 hour Urine	Sodium	SCnc	24H	Urine	Qn		CHEM	mmol/L		Both
	57382-4	Sodium [Moles/volume] in 2 hour Urine	Sodium	SCnc	2H	Urine	Qn		CHEM	mmol/L		Both
	74685-9	Sodium [Moles/volume] in 4 hour Urine	Sodium	SCnc	4H	Urine	Qn		CHEM	umol/L		Both
	94842-2	Sodium [Moles/volume] in Urine from Fetus	Sodium	SCnc	Pt	Urine*Fetus	Qn		CHEM	mmol/L		Observation

2.4.3.2 Qualitative *Properties* and *Scales*

Qualitative *Properties* describe what is being measured for non-quantitative laboratory concepts, such as the presence of an analyte or the specific type of analyte detected. The two most used *Scales* corresponding to qualitative properties are **Ord** (ordinal) and **Nom** (nominal).

The **Ord Scale** is used for categorical results that can be placed in an ordered sequence, such as Detected/Not detected, Low/Medium/High, and Positive/Negative. All tests that are reported in this manner are labeled as ordinal because the result categories can be ranked. The **Ord Scale** is usually associated with the *Property* **PrThr** (Presence or Threshold), meaning either the actual presence or absence of an analyte or that the amount of analyte detected is over some predetermined threshold.

Currently, the **Ord Scale** can also be associated with a range type *Property* such as **NCnc Range** (number concentration range) which reports in a series of ordered “buckets” of values, or the kind of reporting typical in urinalysis such as 1+, 2+, 3+, etc. There has been some discussion about introducing a semi-quantitative *Scale* in LOINC to be a subclass of the **Ord Scale** that would cover observables reported in ranked numerical buckets. However, as of the writing of this guide, this has not been established, and those all remain as **Ord Scale** in LOINC.

The **Nom Scale** is used for categorical results that cannot be ranked, such as the specific *Streptococcus pneumoniae* serotype that is detected, or the species of bacteria identified in a blood culture. The **Nom Scale** is commonly associated with the *Property* **Prid** (Presence or identity), **Type**, or **Imp** (**Impression/Interpretation**).

Prid is used to report the presence of an analyte, and the identity if known. For example, when a bacterial culture is performed, it is not known in advance whether any bacteria will be present. Therefore, one possible result for a bacterial culture is “No growth.” But if one or more types of bacteria are determined to be present, then the specific identity (e.g., *Staphylococcus aureus*) is determined and reported.

Type is used for assays that identify the specific analyte in cases when the baseline presence of the analyte is known. For example, if an isolate known to contain *Streptococcus pneumoniae* is further analyzed to determine the specific serotype, the serotype assay would have a *Property* of **Type**, not **Prid**, because the presence of the *Streptococcus* was already known.

Imp is used for results that represent a diagnostic statement, interpretation, or abstraction of some other observation or group of observations.

KEY POINTS

- *Property PrThr* + *Scale Ord* = presence or absence of an analyte
- *Property Prid* + *Scale Nom* = absence of an analyte, or if present, the type of analyte found; the result is often selected from a list of organisms
- *Property Type* + *Scale Nom* = type of analyte found when the presence is already known; the result is often selected from a list of organisms

2.4.3.3 Qualitative Results vs. Interpreting Quantitative Results

It is important to distinguish qualitative results from an ordinal interpretation of a quantitative result. For example, a routine, rapid test for *Streptococcus pyogenes* antigen yields a true qualitative result (whether the antigen is present) and there is no quantitative result that can be reported. There may be a particular concentration at which the test reads positive, but that information is not part of the reported result. Alternately, an assay for which the concentration of a particular analyte is its primary reportable result is fundamentally a quantitative assay. The quantitative result may be interpreted as positive or negative based on a particular threshold, but the interpretation of the quantitative result, which would be reported in OBX-8 of an HL7 version 2 message, is different from a primary qualitative result, which would be reported in OBX-5.

The package insert will tell you whether a particular assay is approved for quantitative and/or qualitative reporting. Some assays are approved for both and may map to two LOINC codes. For assays that map to two LOINC codes, the code that is appropriate for the result being reported in a particular instance should be used.

KEY POINTS

- Result_value: Positive or Negative → LOINC *Scale* = **Ord**
- Result_Value: 3 mg/dL → LOINC *Scale* = **Qn** (May be interpreted as Positive or Negative, but the interpretation is not the primary result.)

2.4.4 Time Aspect

The *Time Aspect* axis of a LOINC term is usually straightforward for mapping. Most laboratory tests are “point in time” measurements, meaning that they are measuring the amount of analyte present in a given sample at the time the specimen was collected. This is represented by a value of “**Pt**” in the *Time Aspect* of the LOINC term. In other cases, the *Time Aspect* is more distinctive. For example, timed specimen collections, such as 24-hour urine measurements in chemistry, measure the amount of analyte collected in the urine over 24 hours. The *Time Aspect* for such tests is **24H**. While most timed drug and toxicology LOINC terms reflect a 24-hour time frame, there are some for other time periods such as an 8 Hour specimen collection.

If your implementation uses the same result field for random urine and timed urine collections, use the **XXX Timing** LOINC terms for those analytes.

Other LOINC codes commonly used in reporting testing results include for timed urines, “hours of collection,” and “total volume of urine.” Other LOINC codes for “age,” or “body weight,” may also be mapped to test results in information systems that are part of a 24-Hour Urine panel.

2.4.5 System

The LOINC *System* represents the type of sample that is being tested, which could be derived from the patient or the environment. Examples include, but are not limited to, blood (**Bld**), serum or plasma (**Ser/Plas**), urine, various respiratory specimens, cerebrospinal fluid (**CSF**), stool, tissue, pleural fluid, and urethral swab. Testing may also be performed on saliva, meconium, hair, gastric fluid, dried blood spot, or cord blood, such as for newborn drug screening. Environmental specimens, including air and water, may be performed more commonly in toxicology and public health laboratories such as when water is tested for lead.

Forensic testing may be performed on autopsy or other postmortem specimens, such as vitreous fluid or blood taken from the heart. While some laboratories may use their clinical testing methods on postmortem or autopsy specimens, many specifically indicate drug screens on their test menus are not to be utilized for forensic or non-forensic postmortem specimen testing.

Some LOINC terms contain a *System* with multiple specimen types separated by a slash (/). The slash indicates that any of these specimen types are suitable for that test. It does not mean that a specific manufacturer’s assay must be approved for all the specimens listed. It simply means that the results are expected to be clinically equivalent regardless of which of the two (or three) specimens are used and the assay should be approved for at least one of the given specimens.

For instance, the **Ser/Plas** *System* in LOINC is used for assays where the analyte can be measured in either serum or plasma. This is common in antibody tests. LOINC does not make separate terms for tests on serum and plasma unless there is a compelling reason to do so. One such example is ammonia assays, where the preferred specimen is plasma, and the results in serum compared to plasma are significantly different. LOINC will not have a test with a *System* of **Ser/Plas** and the same test but with a *System* of only **Ser** or **Plas**. As testing methods have evolved, by policy, LOINC has added **Plas** to many LOINC terms that originally had a *System* of **Ser**, and we will continue to do so moving forward.

KEY POINTS

- LOINC *System* **Ser/Plas** = “Serum OR Plasma”
- Mapping a particular manufacturer’s assay to a **Ser/Plas** term does not imply that the specific assay is approved for both specimens. Laboratories do not need to run the test on both serum and plasma to use a LOINC code with a *System* of **Ser/Plas**.

LOINC uses a *System* of **XXX** for terms where either the specimen is truly unknown (e.g., a solid or liquid that is not otherwise identified) or is expected to be specified in a field other than the LOINC name. The latter situation is common for testing in microbiology, pathology, genetics, toxicology, and blood fluid tests in cases where a single test can be performed on more than

one specimen type or source. In these situations, the response to an Ask at Order Entry (AOE) question, indicating what type of specimen (e.g., a body fluid) or source (e.g., origin of tissue) was collected is communicated to the performing laboratory. A similar result of “Specimen” or “Source” mapped to its own LOINC is often reported with other test result(s) as part of a panel. In other cases, specimen type is reported separately in the HL7 SPM 4 field (and mapped to a SNOMED CT code from the specimen hierarchy). The specimen source is reported in the HL7 SPM 8 field and mapped to a SNOMED CT code from the anatomic body site hierarchy. Qualifier values may accompany the specimen type in SPM 5 or the source in SPM 9, to indicate further detail, such as post transfusion reaction specimen or “right knee” for specimen source, for example.

The **XXX** terms should only be used in the two cases described above. They should not be used when the test you are mapping is always run on a specific specimen, often designated in the Intended Use section of the package insert for a particular assay. In some cases, mapping to the more general specimen agnostic **XXX** term may be useful, but this practice is discouraged as it results in a loss of information and a failure to accurately map across institutions. (See [Section 2.5.1](#) Special issues related to XXX as a *System* in the LOINC Users’ Guide for more information.)

2.4.6 Method

LOINC includes terms that specify a particular laboratory method as well as those that do not (i.e., methodless terms). In general, when the specific method used has a clinically significant impact on interpreting the results for a particular analyte, LOINC will specify the *Method* in its term name.

Even when the *Method* is included in the term name, LOINC specifies the *Method* at a general level. Where laboratories do not tend to include the method in the name (e.g., most of chemistry) we do not include the *Method* in the name. Where they tend to (e.g., in immunochemistry) we do. For example, Immunoassay (**IA**) includes enzyme immunoassay and chemiluminescence. For some tests, this can be justified by the standardization of methods to produce “equivalent” results, and sometimes by the many variables (method, reagent) that one could never hope to represent fully in a single name. However, even when we do distinguish these cases, we distinguish by type of method, not the most detailed possible method distinction.

The same principle of mapping to the most specific term possible without overspecifying applies to choosing LOINC terms with the appropriate *Method*. If the assay being mapped has a specific methodology important for interpreting the result, then map to a LOINC term with that specific *Method*. If such a term does not exist, a new term may be needed. (See [Section 3.4](#), Requesting new LOINCs.)

2.5 Orders vs. Observations

LOINC terms represent orders, observations, and in many cases, both an order and an observation. In cases where a single orderable test may also be the same test

observation/result, LOINC does not create two different terms. Rather, the designation of whether a term can be used as an order, observation, or both is carried in the OrderObs field within the LOINC database.

IVD test vendors manufacture assays for the determination of an analyte in a specimen. The analyte is part of the laboratory test name for the “result” and its “value” arising from the test analysis. The laboratory test result name may or may not include additional information such as the specimen, test method, and whether the value is qualitative or quantitative, etc., all of which varies depending on the test. Results are often referred to as observations or observables, especially outside of the laboratory.

A single test result can also be orderable, such as a blood lead level. Any order with more than one result reported is referred to as a panel. Orders can be a single orderable, a simple panel, a reflex order, or what’s known as a profile, convenience panel, or tiered order featuring a combination of other single and/or panel orders. Mapping profiles is dependent on how they are structured in the LIS, EHR, or other information system. Some EHRs will offer a “stroke panel” consisting of a PT/INR test, CBC, and BMP as a one-click option for ordering providers. In some cases, the order is a “stroke panel” and given no single LOINC exists, it may not be mappable. In other cases, “stroke panel” is separated out behind the scenes, and the individual CBC, BMP, and PT/INR do have LOINC codes that can be mapped to these orders.

Reflex testing, such as when another order is added on to the specimen based on results of the first test performed on the specimen, is also common, especially in toxicology testing. A Drug Screen order may offer a reflex confirmatory testing order on analytes positive in the screening test. Like profiles, LOINC codes are not available for many reflex orders but are available for many drug orders such as screening and confirmatory panels.

For example, *Multiplex allergen panel* is an order, but it is not an observation because it does not represent a single test that can be resultated. The OrderObs value for such a term is **Order**. On the other hand, *European pigeon tick IgE Ab* is a single test that could be ordered and resultated individually, so in LOINC, its OrderObs value is **Both** (meaning that the same term can function as both the order and observation). Finally, some observations are such that they would never be ordered by themselves, and these have an OrderObs value of **Observation**, such as *Pine IgE Ab RAST Class*.

The concepts of orders and observations are represented by the panel and panel children constructs in LOINC. LOINC panels are always order-only terms that contain a discrete set of enumerated child elements (children), corresponding to what is typically called a “battery” or “profile” in the laboratory setting. A given panel term does not represent a single resultable test. Rather, it represents an order for a set of tests. Panel terms often have a dash (-) for the *Property* and *Scale* since the child terms may not all have the same values for these attributes. While in almost all cases the child elements within a LOINC panel are enumerated, they are not necessarily all required. Some may be optional or conditional, and the same LOINC panel term may be used by different laboratories or manufacturers whose specific batteries may not be exactly the same. This is especially true with drug panels. Some laboratories have over a hundred panels on their menu with different combinations of drugs tested therein. LOINC codes may

never exist for each combination of drugs, especially with all the synthetic and illicit drugs on the market.

LOINC terms that are observations only are, by definition, children, or resultable tests within an orderable panel, and cannot be ordered by themselves. In contrast, those classified as both an order and an observation may be included in a panel, but also must be orderable (and resultable) on their own.

For more information regarding panels and business rules for mapping local batteries to LOINC panels, see [Section 7](#) Panels (Batteries), and [Section 12.1](#) Business rules for users mapping their local panels to LOINC in the LOINC Users' Guide.

KEY POINTS

- Individual observations should only be mapped to LOINC codes that have an *OrderObs* value of **Observation** or **Both**
- Batteries or panels should only be mapped to LOINC codes that have an *OrderObs* value of **Order**
- Not all laboratory information systems utilize Order LOINC. In such situations, only **Observation** or **Both** would be used at the test level.

3 Efficient Searching

There are many strategies for searching in LOINC. Some are more suited for experienced users and some for beginners. But they all can be effective if done properly and systematically.

3.1 Searching for Beginners

For LOINC beginners, or those unfamiliar with the range of content it contains, we recommend starting at a high level with the *Component* and possibly one other key piece of information, such as the *System* or Units, if applicable, or *Method*, if available. From there you can narrow your search results. This helps gain an understanding of how different lab tests are represented in LOINC. It also helps determine what additional information might be needed before choosing a LOINC code, especially if more than one choice exists after your initial search.

For example, just your lab test name of “Aspergillus Ab” may seem like enough information. However, searching for Aspergillus Ab in LOINC will return 169 candidate terms, as shown below in the upper left Results tab. Only a few possible candidates are shown below.

LOINC		Aspergillus Ab		Search	
RESULTS		DISPLAYING	200	FILTER	VIEW
169		1-169		List	Card
Status		EXPORT			
LOINC	Long Common Name	Component	Property	Timing	System
Scale	Method	Class	Type	Example UCUM Units	Order/ Observation
7807-1	Aspergillus sp Ab [Units/volume] in Serum	Aspergillus sp Ab	ACnc	Pt	Ser
7808-9	Aspergillus sp Ab [Presence] in Serum	Aspergillus sp Ab	PrThr	Pt	Ser
22087-1	Aspergillus sp Ab [Titer] in Serum	Aspergillus sp Ab	Titr	Pt	Ser
33266-8	Aspergillus sp Ab [Titer] in Urine	Aspergillus sp Ab	Titr	Pt	Urine
31233-0	Aspergillus sp Ab [Units/volume] in Cerebral spinal fluid	Aspergillus sp Ab	ACnc	Pt	CSF
42766-6	Aspergillus sp Ab [Interpretation] in Serum by Immune diffusion (ID) Narrative	Aspergillus sp Ab	Imp	Pt	Ser
18426-7	Aspergillus sp Ab [Presence] in Cerebral spinal fluid	Aspergillus sp Ab	PrThr	Pt	CSF
43870-5	Aspergillus sp Ab [Presence] in Serum by Complement fixation	Aspergillus sp Ab	PrThr	Pt	Ser
5052-6	Aspergillus sp Ab [Presence] in Serum by Immune diffusion (ID)	Aspergillus sp Ab	PrThr	Pt	Ser

Right away, you can see that you need more information from the laboratory. What species of *Aspergillus* is tested? Is an IgA, IgE, or IgG antibody tested? Is the result quantitative or qualitative, or perhaps a titer? If quantitative, what units are reported? Is it a microbiology test or an allergy test? Are you mapping an order panel or a single result item? Is it a RAST test?

What if the test name is “*Aspergillus fumigatus*+*Aspergillus niger* IgE Ab”? You could search for all those key words and narrow down your candidate terms right away. In addition, looking at the range of existing terms is helpful in deciding whether a new term should be requested if a specific match is not found.

3.2 Experienced Searchers

If you are more familiar with LOINC and the types of terms that it contains, we recommend focusing your search from the start and including all key information about *Component*, *Timing*, *Property*, *System*, and *Method* in the search string. If you need a LOINC code for a test named “Peanut Recombinant IgG4” testing for peanut recombinant (rAra h) 1 IgG4 Ab reported in mg/L, include all that information in the search. If you search for “peanut” by itself, you will be faced with numerous LOINC terms, but searching for “peanut recombinant” narrows the list to 19. Adding IgG4 limits it to six LOINC codes. Alternatively, adding in the *Property*, which you know is **MCnc** based on the units of measure, and searching for “peanut recombinant MCnc” returns six LOINC codes.

Specific details, for example searching for “(rAra h) 1,” help narrow LOINC results more quickly. Typos or terms that are too specific may return zero matches, so enter search terms carefully. As you work with different types of laboratory tests, you’ll become more familiar with which details help return LOINC matches most quickly.

There are many other fields in addition to those mentioned here that can be used to focus searches. For example, if you are only looking for new terms released for the first time in the 2.72 LOINC release, you can use the VersionFirstReleased field, i.e., “VersionFirstReleased:2.72,” as well as other search parameters. Refer to the [LOINC search help](#) for a full list of searchable fields and search syntax.

3.3 Synonymy in LOINC

A search for LOINC terms in RELMA or the web tool (search.loinc.org) will detect not only words in the fully specified name, but also examine synonyms for each of the LOINC parts that make up that LOINC term. For each LOINC term, these synonyms are assembled into a list called “Related Names.” As an example, a search for a brand name drug such as “Vicodin” will detect LOINC terms for terms with *Components* of HYDROcodone and oxyCODONE . Likewise, a search for elemental cyanide, “CN,” would reveal terms for cyanide, even though “CN” does not appear in the *Component* or any of the term names. The “CN” search also returns other related terms such as “cranial nerve” which is not applicable to laboratory test mapping. While the term “lead,” with its distinct pronunciations of “led” for the lab test, and “leed” (lēd) for the EKG device provides the appropriate context and meaning in speaking, more information is needed for the computer to narrow its search to the appropriate context and meaning.

Synonyms in LOINC will often facilitate detection of likely possible terms, but LOINC’s related names are not always equivalents. When a search detects a LOINC term through related names it does not mean that term represents exact equivalence to the search item. For example, the search phrase “ABG” (for arterial blood gas) will detect the sodium term ABG, even though arterial blood gas is not equivalent to a sodium level. It is detected because “ABG” is linked as a related name to the *System* **BldA** (arterial blood) and is reasonable because sodium can be reported by blood gas analyzers. LOINC’s related names will help to increase the detection rate when searching for terms, however, a certain degree of domain expertise is required to confirm whether the relationship is equivalent.

3.4 Using LOINC Panels to Find Individual Test Codes

While mapping panels and lab orders is out of scope for this guide, it is worth noting that mapping the panel first can be a helpful start to mapping the associated results. Be careful as you must still be sure the individual terms match on all the *Part* values.

The basic and comprehensive metabolic panels are some of the most common tests performed. Clinicians routinely order these tests by panel rather than individually, for example, the lab order will be for a respiratory allergen panel, US - Hawaii panel rather than individual orders for cat dander, dog dander, American house dust mite, etc. Insurance and payment regulations may require some tests to be ordered as a panel for payment, while other tests may be required to be ordered individually depending on diagnostic coding at order entry.

KEY POINTS

- New LOINC users should start with broad searches, then narrow results.
- Experienced LOINC users should be very specific to begin their search.
- Using related names can be helpful but should not be accepted without further review because related does not equal equivalent.
- Panels can help lead searchers to single test codes.

4 Mapping an Assay Approved for Multiple Specimens and Resulting Styles

Assays approved for use with multiple specimens and multiple reporting options will map to multiple LOINC codes. Therefore, mapping from an IVD vendor’s perspective is different from that of a laboratory end user. When vendors provide a list of LOINC codes to their clients, they should include all the appropriate codes based on the information in their package insert. Laboratories using vendor mappings should then pick the subset of codes applicable to their local implementation of the assay.

The same mapping principles apply to each approved specimen and reporting format. For example, if an assay is approved for use with either serum or CSF, the vendor should provide two LOINC codes – one with *System* **Ser** or **Ser/Plas**, and one with **CSF**. Similarly, if a single assay can be configured to result in an ordinal qualitative response or a numeric titer, the vendor should provide two different LOINC codes – one for each type of result. It is up to each downstream client to determine the best LOINC codes for their specific implementation of the test.

Note: A lab may decide to follow another mapping strategy, as shown for “Laboratory C” in the table below. Take note of the limitation presented in the footnote for that strategy.

KEY POINTS

- IVD vendors should map to all the LOINC codes that support the range of specimens and reporting options approved for a single test kit.
- Laboratories should map to the subset of LOINC codes appropriate for the specimens and reporting options they have implemented for that test kit.

The following table illustrates possible sets of LOINC codes IVD vendors could provide for various hypothetical assays. They are based on package insert information and the subset(s) of codes downstream laboratories would use based on their individual LIS implementations. The example is from the Micro domain, but the same principles apply to almost all domains.

Package insert assay information	IVD Vendor provided LOINC codes	LOINC codes used by downstream clients for individual implementations
Adenovirus DNA PCR • Approved for serum, CSF, and urine • May only be reported as a qualitative ordinal result	21055-9 Adenovirus DNA: PrThr:Pt: Ser/Plas: Ord: Probe.amp.tar 38375-2 Adenovirus DNA: PrThr: Pt:CSF:Ord: Probe.amp.tar 86512-1 Adenovirus DNA: PrThr: Pt:Urine:Ord: Probe.amp.tar	Laboratory A: only using the assay to test CSF specimens 38375-2 Adenovirus DNA: PrThr: Pt: CSF:Ord: Probe.amp.tar Laboratory B: using the assay for all three approved specimens, with three individual test builds 21055-9 Adenovirus DNA:PrThr:Pt: Ser/Plas:Ord:Probe.amp.tar 38375-2 Adenovirus DNA: PrThr: Pt: CSF: Ord: Probe.amp.tar 86512-1 Adenovirus DNA: PrThr: Pt:Urine: Ord: Probe.amp.tar Laboratory C: using the assay for all three specimens, but with a single build in the LIS that does not specify the specimen² 39528-5 Adenovirus DNA: PrThr:Pt: XXX: Ord: Probe.amp.tar
Adenovirus DNA PCR • Approved for CSF only • May be reported as a qualitative ordinal result or a quantitative viral load as copies/volume or log copies/volume	38375-2 Adenovirus DNA: PrThr: Pt:CSF:Ord:Probe.amp.tar 49338-7 Adenovirus DNA: NCnc: Pt:CSF:Qn:Probe.amp.tar 66730-3 Adenovirus DNA: LnCnc: Pt:CSF:Qn:Probe.amp.tar	Laboratory D: result built in the LIS as an ordinal text string 38375-2 Adenovirus DNA:PrThr: Pt: CSF:Ord:Probe.amp.tar Laboratory E: numeric result expressed as # copies/mL 49338-7 Adenovirus DNA:NCnc:Pt: CSF:Qn:Probe.amp.tar Laboratory F: numeric result expressed as log copies/mL 66730-3 Adenovirus DNA:LnCnc:Pt: CSF:Qn: Probe.amp.tar Laboratory G: will report out all three results, so each result will have the corresponding LOINC seen above.

² Using a single build for a generic specimen is not recommended as it results in a loss of information, and the list of codes provided by the vendor should not include the generic LOINC code when more specific ones are available (note that [39528-5](#) is not listed in the second column).

Adenovirus DNA PCR • Approved for serum, CSF, and urine • May be reported as a qualitative ordinal result, quantitative viral load in copies/volume or log copies/volume, or both an ordinal result and if positive, a quantitative viral load	21055-9 Adenovirus DNA: PrThr:Pt: Ser/Plas: Ord: Probe.amp.tar 38375-2 Adenovirus DNA: PrThr: Pt: CSF: Ord: Probe.amp.tar 86512-1 Adenovirus DNA: PrThr: Pt: Urine: Ord: Probe.amp.tar 49334-6 Adenovirus DNA: NCnc: Pt: Ser/Plas: Qn: Probe.amp.tar 49338-7 Adenovirus DNA: NCnc: Pt: CSF: Qn: Probe.amp.tar 48609-2 Adenovirus DNA: NCnc: Pt: Urine: Qn: Probe.amp.tar 66723-8 Adenovirus DNA: LnCnc: Pt: Ser/Plas: Qn: Probe.amp.tar 66730-3 Adenovirus DNA: LnCnc: Pt: CSF: Qn: Probe.amp.tar 66719-6 Adenovirus DNA: LnCnc: Pt: Urine: Qn: Probe.amp.tar	Laboratory H: only using the assay to test CSF specimens, reporting results as a qualitative ordinal 38375-2 Adenovirus DNA: PrThr: Pt:CSF: Ord: Probe.amp.tar Laboratory I: testing CSF and urine specimens, reporting results as copies/mL 49338-7 Adenovirus DNA: NCnc: Pt: CSF: Qn: Probe.amp.tar 48609-2 Adenovirus DNA: NCnc: Pt:Urine: Qn: Probe.amp.tar Laboratory J: testing all three specimens, reporting initial results as ordinal, followed by viral load in log copies/mL if the ordinal result is “Detected” 21055-9 Adenovirus DNA: PrThr: Pt: Ser/Plas: Ord: Probe.amp.tar 38375-2 Adenovirus DNA: PrThr: Pt:CSF: Ord: Probe.amp.tar 86512-1 Adenovirus DNA: PrThr: Pt:Urine: Ord: Probe.amp.tar 66723-8 Adenovirus DNA: LnCnc: Pt: Ser/Plas: Qn: Probe.amp.tar 66730-3 Adenovirus DNA: LnCnc: Pt: CSF: Qn: Probe.amp.tar 66719-6 Adenovirus DNA: LnCnc: Pt: Urine: Qn: Probe.amp.tar
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5 Mapping Qualitative Result Values

Qualitative results are non-numeric values that can be either ordinal or nominal. Ordinal result values can be ranked, such as Detected/Not detected, Low/Medium/High, Susceptible/Intermediate/Resistant, or Positive/Negative. Nominal result values cannot be ranked and include concepts such as the specific *Streptococcus pneumoniae* serotype detected, the species of bacteria identified in a blood culture, or the specific cell markers detected. Nominal result values can be thought of as “short answers” or those coming from a picklist. Narrative result values (the *Scale Narr* in LOINC) are longer, free-text entries that are not drawn from a formal vocabulary or terminology standard.

Qualitative results also include laboratorian reviews of quantitative and/or non-numeric values. These reviews or interpretations are covered in LOINC by either role or department specifications. They can be found by searching for “review xxx” in RELMA. Role reviews include supervisor, medical director, and pathologist reviews. Department reviews include clinical biochemist, endocrinologist, Flow cytometry, and more.

LOINC

review XXX

Search

RESULTS

14

DISPLAYING 1-14

200

FILTER

VIEW

EXPORT

List

Card

Status

LOINC	Long Common Name	Component	Property	Timing	System	Scale	Method	Class	Type	Example UCUM Units	Order/Observation
69047-9	Geneticist review of results	Geneticist review	Imp	Pt	XXX	Nar		MISC			Observation
69427-3	Supervisor review of results	Supervisor review	Imp	Pt	XXX	Nom		MISC			Observation
59464-8	Microbiologist review of results	Microbiologist review	Imp	Pt	XXX	Nar		MICRO			Observation
69053-7	Endocrinologist review of results	Endocrinologist review	Imp	Pt	XXX	Nar		MISC			Observation
69048-7	Immunologist review of results	Immunologist review	Imp	Pt	XXX	Nar		MISC			Observation
59466-3	Hematologist review of results	Hematologist review	Imp	Pt	XXX	Nar		HEM/BC			Both
69051-1	Urinalysis specialist review of results	Urinalysis specialist review	Imp	Pt	XXX	Nar		MISC			Observation

In the context of HL7 v2 data exchange, both the observation and the observation result value(s) are reported in the OBX segment, with the observation in OBX-3 and the result value(s) in OBX-5. More specifically, OBX-5.1 contains the code, 5.2 the text, and 5.3 the coding system. In cases where the *Property* is **OrdQn**, such as antimicrobial susceptibility testing, result values can be reported as either:

- a quantitative value in OBX-5 with an interpretation in OBX-8; or
- an ordinal result value in OBX-5 using the same LOINC code.

This section addresses the second reporting style, however, example OBX segments for both are shown here for comparison.

OBX|1|NM|6932-8^Penicillin [Susceptibility] by Minimum inhibitory concentration (MIC)^LN||0.05|ug/mL||S||F|...

OBX|1|CE|6932-8^Penicillin [Susceptibility] by Minimum inhibitory concentration (MIC)^LN||LA6576-8^Susceptible^LN|||||F|...

Example HL7 v2 OBX segments are distinguished in a bordered box. HL7 v2 examples, rather than CDA or FHIR examples, are used given their wide use in laboratory result reporting at the time this guide was written. Note that these examples are simplified for readability and do not represent the full OBX structure. For more information, refer to appropriate HL7 v2 and related specifications. The LOINC code in each example links to the web page with details for that term.

The LOINC distribution includes structured Answer Lists of possible result values for nearly every ordinal qualitative LOINC observation term as well as many nominal terms. These LOINC Answer Lists are sets that organize a collection of LOINC Answer strings and codes. LOINC Answer Lists illustrate the possible result values associated with an observation, and LOINC Answer codes provide unique identifiers for the strings representing them. A specific LOINC Answer List may be associated with one or more individual LOINC terms. Please note that LOINC Answer lists (“LL” codes) and Answer strings (“LA” codes) are not managed in LOINC as distinct terminologies, but rather are defined only in the context of regular LOINC codes. This means that LL and LA codes will only be found in a LOINC release if they are associated with a fully specified LOINC code in that release.

KEY POINT

- Local result values for qualitative tests can be mapped to LOINC Answer codes representing the equivalent string.

Different IVD manufacturers may recommend different result values, such as Detected/Not detected versus Present/Absent, for similar assays. Always follow local reporting regulations. For example, report qualitative results according to the IVD vendor's package insert for the assay used. Ongoing efforts in the U.S. by the FDA's Systemic Harmonization and Interoperability Enhancement for Lab Data (SHIELD) initiative, the IVD Industry Connectivity Consortium (IICC).

5.1 Linking LOINC Answer Lists to LOINC Terms

In general, laboratory terms with Answer Lists use **Example** type lists. The intent is to give users an idea of the types of results associated with a particular LOINC observation. For example, a list that contains result values such as "Detected" and "Not detected" is indicative of an ordinal result representing the presence or absence of an analyte. **Example** lists are illustrative of possible result values, or a "starter set," to which additions or subtractions can be made depending on the use case. **Preferred Example** lists contain a set of answers that users are strongly encouraged to use. They represent a recommended set however alternate values may be used if necessary. **Normative Example** lists are typically defined by a validated survey questionnaire (e.g., patient reported outcomes, instruments), government form, or other authoritative source. When using LOINC terms bound to **Normative** Answer Lists, only answers in the specified set are allowed. For laboratory terms, both **Normative** and **Preferred** Answer Lists are relatively uncommon.

A LOINC term is typically bound to an Answer List based on information from the package insert and sample report(s) of the original test the term was created for. When new LOINC terms are requested, we ask for supporting documentation, in part to understand how the observation is reported. If such documentation is not provided, we work with the submitter to determine the appropriate set of result values for a particular assay.

KEY POINTS

- **Example** Answer Lists include Answer codes and strings that are meant to be illustrative of possible result values.
- Answer Lists are typically bound to a particular LOINC term based on the IVD test upon which that term was originally modeled.
- Depending on their reporting needs, users may add or subtract result values from an **Example** LOINC Answer List provided they remain consistent with that term's *Scale* (i.e., ordinal or nominal).

5.2 LOINC Answers and SNOMED CT Concepts

SNOMED CT is an international terminology standard for clinical information, such as symptoms, diagnoses, anatomy, and microorganism names³. SNOMED CT provides codes representing nominal and ordinal values for observations represented by LOINC terms and that are organized by hierarchies, such as Clinical finding, Organism, and Qualifier value. If appropriate concepts exist in SNOMED CT, the Regenstrief Institute recommends using SNOMED CT codes for reporting qualitative laboratory result values if appropriate licensure is granted. In situations where the use of SNOMED CT is not permitted, LOINC Answer codes or other terminologies could be used.

Some LOINC Answer Lists include a mapping between the LOINC Answer codes and corresponding SNOMED CT codes. This is ongoing work and a given list may only contain mappings for a subset of the answers it contains. Mappings are also shown on detailed display pages associated with each term.

Note: Due to licensing restrictions, this guide does not include SNOMED CT codes. Mappings are available in the LOINC Answer File, available for [download from the LOINC website](#).

5.3 Ordinal Result Values

LOINC contains a variety of Answer Lists that include sets of ordinal result values. The three LOINC Answer Lists shown in the table below are the most common lists attached to LOINC laboratory terms other than those terms in the **ALLERGY Class**. These three lists account for 75% of the non-allergy terms that have ordinal lists. In the **ALLERGY Class**, more than 90% of ordinal terms are linked to a single Answer list, [LL1927-4](#), which represents the results of RAST testing. The rest are linked to [LL360-9](#).

Answer List ID	Answer string	Answer code
LL744-4	Detected	LA11882-0
	Not detected	LA11883-8
LL360-9	Positive	LA6576-8
	Negative	LA6577-6
LL1937-3	Present	LA9633-4
	Absent	LA9634-2

Result value mapping should be done at the level of the Answer code and Answer string, not at the level of the Answer List. For example, if an IVD manufacturer recommends reporting result values as either Present or Absent, you would map the Present result value to LA9633-4 and the Absent result value to LA9634-2; nothing would be mapped to Answer list [LL1937-3](#).

³ SNOMED CT [Internet]. London (England): SNOMED International. Available from: <http://www.snomed.org/snomed-ct>

Similarly, if the recommended result values are Positive and Absent, and there is no LOINC Answer List that bundles those two together, you would still map to the individual answers LA6576-8 and LA9634-2.

KEY POINT

- Individual result values should be mapped to the appropriate Answer code and Answer string pairs, not directly to an Answer List.

Also, recall that the possible result values for a particular assay are defined by the product information for that assay and not by the list associated with the corresponding LOINC term. For instance, suppose two different assays, “Assay A” and “Assay B,” produce ordinal result values for whether IgM antibody to *Legionella* species is present in a serum specimen. The package insert for Assay A recommends reporting result values as Detected/Not detected, whereas the package insert for Assay B recommends reporting Present/Equivocal/Absent. Laboratories using Assay A would use Answer codes LA11882-0 (Detected) and LA11883-8 (Not detected), and laboratories using Assay B would use LA9633-4 (Present), LA11885-3 (Equivocal), and LA9634-2 (Absent). Both laboratories would map to the same LOINC term.

[81118-2](#) *Legionella* sp Ab.IgM:PrThr:Pt:Ser:Ord
LCN: *Legionella* sp IgM Ab [Presence] in Serum

The above LOINC term happens to be associated with **Example** LOINC Answer List [LL744-4](#), which contains LA11882-0 (Detected) and LA11883-8 (Not detected), but the same term can be used with result values LA9633-4 (Present), LA11885-3 (Equivocal), and LA9634-2 (Absent), or any other ordinal values that have the same meaning.

KEY POINTS

- Primary characteristics of an assay should determine which LOINC term to map to, not the Answer List associated with that term.
- Result value mapping is an independent process.

5.4 Nominal Result Values

Nominal results include concepts such as the type of microorganism, drug, protein, cell, etc. identified in a specimen. LOINC contains many more nominal than ordinal lists due to the variety of possible nominal result values across LOINC terms. However, the general principles of using LOINC Answer Lists are the same. **Example** Answer Lists are representative of the types of results associated with a given observation. The primary observation should be mapped to a LOINC term first, followed by mapping the allowable result values.

One difference between nominal and ordinal results is that LOINC Answer strings and codes may not exist for all reportable nominal results given the breadth of result values possible. If no standard code for a nominal laboratory result value is available, whether from LOINC, SNOMED CT, or another coding system, it is possible to send a free text entry as the result value (e.g., in

an HL7v2 OBX segment). However, such unstandardized reporting makes it difficult for receiving systems to aggregate and process the data and is strongly discouraged.

5.5 Uncertain Result Values

For both ordinal and nominal types of results, there are instances in which a test may not produce a definitive result value. In many cases, IVD manufacturers specify the value that should be used in various circumstances. LOINC contains a variety of Answer codes and strings that represent these uncertain types of results as shown below.

Answer string	Answer code
Equivocal	LA11885-3
Indeterminate	LA11884-6
Inconclusive	LA9663-1
Invalid	LA15841-2
Undetermined	LA14100-4
Uncertain	LA12719-3
Uncertain significance	LA26333-7

5.6 Flavors of Null

LOINC does not make different Answer Lists that vary only by the inclusion or exclusion of certain null flavors. Our policy is to add specific null flavors as entries in the Answer List only where we are aware of their specific relevance. However, LOINC Answer codes exist for a variety of null flavor concepts. Users wishing to communicate such a null flavor concept are free to do so, regardless of its presence in the LOINC Answer List. Examples of LOINC Answer strings and codes for null flavor result values are shown in the table here.

Answer string	Answer code
Unknown	LA4489-6
Not done	LA6630-3
Test not done	LA18373-3
Not performed	LA7304-4
Equipment failure	LA7497-6

Part 2: LOINC Mapping Guide for Allergy

1 Introduction

Part one of this guide explained the general concepts and rules of LOINC mapping. The following sections will focus on Allergy-specific information. Be sure to refer to previous sections for term definitions and explanations.

Most LOINC terms used for mapping to allergy laboratory orders and results are in the “Allergy” and “Panels.Allergy” LOINC Classes. The LOINC Allergy domain contains a variety of terms in the following areas: food, pollen, grass, tree, insect, animal, fungal, occupational, inhalant, and respiratory allergens. Allergy testing of these items ranges from Radioallergosorbent Test (RAST) testing to Immunoglobulin E (IgE) antibody testing, to allergen mixes, some of which are regional formulations.

This guide does not cover each of these in detail because, for most allergy tests, the principles presented in part one include the information necessary to choose the correct LOINC term. Instead, this section will show patterns, point out exceptions, and highlight potential problem areas in the allergy domain.

2 General Rules for Naming Components

In the allergy domain, the *Component* is often very straightforward and includes the allergen name plus the type of antibody, such as **Latex Ab.IgE**. However, there are cases where the allergen name may be difficult to find in LOINC due to differences in the commonly used name, or a single common name being used for multiple species.

2.1 Latin Names, Common Names, and Synonyms

The *Component (Analyte)* for an allergen contains the name of the biologic organism that is the source of the allergen. The formal name will use the Latin taxonomic name (e.g., *Arachis hypogea*). The *Long Common Name* will use the common name of that entity (e.g., Peanut) if one exists. In addition, most LOINC Parts have associated synonyms that are assembled into a list called “Related Names.”

LOINC		6206-7							
RESULTS	1	DISPLAYING	200	FILTER	VIEW	EXPORT			
		1-1			List Card				
Status	LOINC	Long Common Name	Component	Property	Timing	System	Scale	Method	Class
	6206-7	Peanut IgE Ab [Units/volume] in Serum	Arachis hypogaea Ab.IgE	ACnc	Pt	Ser	Qn		ALLERGY

For well specified allergens, the *Component* will also indicate whether the allergen has a natural source or has been generated via recombinant method by adding the word “native” or “recombinant,” respectively. Different antigens from the same species are distinguished by the addition of a sequence number that roughly reflects when they were identified.

LOINC		58779-0							
RESULTS	1	DISPLAYING	200	FILTER	VIEW	EXPORT			
		1-1			List Card				
Status	LOINC	Long Common Name	Component	Property	Timing	System	Scale	Method	Class
	58779-0	Peanut recombinant (rAra h) 1 IgE Ab [Units/volume] in Serum	Arachis hypogaea recombinant (rAra h) 1 Ab.IgE	ACnc	Pt	Ser	Qn		ALLERGY

So, you will see names such as the following:

- Arachis hypogaea recombinant 1
- Arachis hypogaea recombinant 3
- Arachis hypogaea native 1
- Arachis hypogaea native 3

As with all LOINC domains, synonyms are also used parenthetically within the allergen *Component* to help users recognize the term. We use standard acronym names, used by most allergen manufacturers, as synonyms. These begin with lowercase “n” or “r” to distinguish native from recombinant allergens if appropriate, followed by the first 3 letters of the genus (in Latin), a space, and the first letter of the species (in Latin). If the first 3 letters of the genus and the first letter of the species are not enough to distinguish between two allergens, the second letter of the species name is added: for example, “Prunus avium recombinant” is “(rPru av) 1.” Because the content in the parentheses represents the antigen acronym and not the Pharmacia code, the antigen sequence number is purposely placed after the closing parenthesis. Finally, Ab.IgE is appended to the end of the *Component* name. (Note that most allergy testing is IgE rather than IgG.)

A search for LOINC terms in RELMA or the web tool (search.loinc.org) will list not only words in the fully specified name but also the display names and synonyms for each of the LOINC parts comprising the LOINC term. For this reason, a search for the common name “Peanut” reveals

LOINC terms with *Components* containing “Arachis hypogaea” because the LOINC part [LP13943-3](#), Arachis hypogaea, has “Peanut” as a display name as well as a synonym.

Putting this all together, the formal and common names of the *Components* for the allergen examples above become:

LOINC code	Formal name of <i>Component</i>	Long common name of <i>Component</i>
58779-0	Arachis hypogaea recombinant (rAra h) 1 Ab.IgE	Peanut recombinant (rAra h) 1 IgE Ab
58777-4	Arachis hypogaea recombinant (rAra h) 3 Ab.IgE	Peanut recombinant (rAra h) 3 IgE Ab
65769-2	Arachis hypogaea native (nAra h) 1 Ab.IgE	Peanut native (nAra h) 1 IgE Ab
65771-8	Arachis hypogaea native (nAra h) 3 Ab.IgE	Peanut native (nAra h) 3 IgE Ab

Keep in mind that related names are exactly as described; they are related to the part to which they are linked and are NOT always equivalent. They may include associated diseases, common variants or previous names, related codes, and even common misspellings. Related names help increase the likelihood that searches return desired terms, however, a certain degree of domain expertise is required to confirm whether the relationship is equivalent.

2.2 Allergen Codes

Many allergens and allergen mixes can be identified with special character codes that most allergy diagnostics vendors reference in their test catalogs. Historically, these codes were assigned primarily by Phadia AB⁴ (formerly Pharmacia AB). Currently, they may be assigned by various manufacturers. These codes typically begin with “f” for food allergens, “t” for trees, “g” for grasses, “fp” or “fm” for food panel or food mix, etc.

Allergen codes are included as related names linked to the appropriate LOINC *Components*. Codes may not be unique across manufacturers, and for allergens that have multiple codes, all the relevant codes get listed as we learn of them. Examples of various allergens and their corresponding allergen codes included as related names are listed below:

- Egg whole
 - Related name: f245
- Egg yolk
 - Related name: f75
- Egg white
 - Related name: f1
- Magnolia grandiflora
 - Related name: t216
- Paspalum notatum (Bahia grass)
 - Related name: g17

⁴ ImmunoCAP Allergens [Internet]. Uppsala(Sweden): ImmunoCAP; c2006 [updated 2007 November 2; cited 2008 June]. Available from: <http://www.phadia.com/en-GB/5/Products/ImmunoCAP-Allergens/>

2.3 Allergen Mixes

Frequently, allergens are tested in combination, where the results do not distinguish between the allergens. Different manufacturers call these allergen mixes or panels. Be careful, as the manufacturer's use of "panel" refers to a combination of allergens all tested together, which is different from the definition of "panel" in LOINC where panels include separate terms for multiple test results.

To clearly represent combinations, the set of allergens tested are enclosed in parenthesis in the LOINC *Component* with the allergens separated by a '+'. In addition, the *Component* display name (which is used in the Long Common Name) includes the words "Allergen Mix" between the type of allergen and the allergen mix number, as shown in the examples below:

(Alternaria alternata + Aspergillus fumigatus + Rye pollen + Wheat pollen) Ab.IgE
Component display name: Occupational Allergen Mix 3 (Alternaria alternata + Aspergillus fumigatus + rye pollen + wheat pollen) IgE Ab

(Ananas comosus + Citrus limon + Fragaria vesca + Pyrus communis) Ab.IgE
Component display name: Food Allergen Mix 16 (Pineapple + Lemon + Strawberry + Pear) IgE Ab

Note that the allergen mix numbers do not indicate the number of allergens included in the test. They represent the manufacturer's code for the particular mix, NOT the number of allergens in the mix. In most cases where the number is greater than 5, this will be obvious, but in cases with smaller numbers, such as the Occupational Allergen Mix 3 example above, it is not as clear.

In general, where allergy testing manufacturers have identical allergen combinations, they use the same number code to identify the mix. However, we know of at least one case in which the same number code is used by different manufacturers for unique food combinations, though with a distinct prefix ("fp" versus "fm"). Phadia's Food Mix 73 contains beef, chicken, and pork, while Siemen's Food Panel 73 contains beef, chicken, pork, and lamb. For this case, and those like it, we will use the same convention of including the prefix in the *Component* display name as well as the number:

(Beef+Chicken+Pork) Ab.IgE
Component display name: Food Allergen Mix fx73 (Beef+Chicken+Pork) IgE Ab

(Beef+Chicken+Pork+Lamb) Ab.IgE
Component display name: Food Allergen Mix fp73 (Beef+Chicken+Pork+Lamb) IgE Ab

KEY POINT

- The number included in the display names for allergen mixes represents the manufacturer's code for the particular mix, NOT the number of allergens in the mix.

2.4 Total

LOINC will avoid use of the word “total” in laboratory test names, except when denoting the denominator of a fraction. Thus, we use “Papain Ab.IgE” not “Papain Ab.IgE.total”, when looking for total Papain IgE in a specimen. However, when looking for the percent of Papain IgE in relation to total IgE, then “IgE.total” is used in the Divisor:

Papain Ab.IgE/IgE.total: AFr: Pt: Ser: Qn

2.5 Plants With Multiple Types of Allergens

Some plants have more than one type of allergen. For example, IgE antibodies can develop against tree pollen and the fruit of the same tree. These different types of allergens have different antibody tests and different allergen codes. Similarly, antibodies exist for grain and grain pollen. In these cases, the LOINC *Component* will contain the word “pollen” to distinguish the pollen allergen from the food allergen. For example, LOINC *Components* and related names for corn (maize) and corn pollen IgE antibody are:

Zea mays Ab.IgE
Related names: f8; cultivated corn; maize

Zea mays pollen Ab.IgE
Related names: g202; cultivated corn; maize

3 Property and Scale

As mentioned, *Scale* can often help quickly narrow the field of candidate LOINC terms; however, 99.9% of allergy terms have a *Scale* of Qn (Quantitative) or Ord (Ordinal). This means that determining the correct *Property* is somewhat easier. Likewise, three LOINC *Properties*, **ACnc** (Arbitrary concentration), **PrThr** (Presence or threshold), and **MCnc** (Mass concentration) account for 93%, with just five other properties comprising the remaining 7% of allergy LOINC terms.

3.1 Properties and Units of Measure

Quantitative LOINC *Properties*, namely **ACnc** and **MCnc**, are tightly related to the units of measure associated with a result, which is why knowing the units of measure helps in choosing the correct LOINC term.

ACnc terms typically have units like kU/L or IU/mL, or some measurement variation thereof. Naturally, **MCnc** terms lead with a mass unit and are divided by a unit of volume, such as ug/mL or pg/mL.

Suppose your laboratory offers a test for Botrytis cinerea Ab.IgG measured in ug/mL. If you search for “Botrytis cinerea Ab.IgG” you will find two candidate LOINC terms, with different *Properties*:

LOINC		Botrytis cinerea Ab.IgG										Search				
RESULTS		2		DISPLAYING		200		FILTER		VIEW		EXPORT				
		1-2		 				List		Card						
Status	LOINC	Long Common Name		Component		Property	Timing	System	Scale	Method	Class	Type	Example UCUM Units	Order/Observation	Version First Released	Version Last Changed
	30022-8	Botrytis cinerea IgG Ab [Units/volume] in Serum		Botrytis cinerea Ab.IgG		ACnc	Pt	Ser	Qn		ALLERGY	 k[arb'U]/L	Both	2.04	2.68	
	42591-8	Botrytis cinerea IgG Ab [Mass/volume] in Serum		Botrytis cinerea Ab.IgG		MCnc	Pt	Ser	Qn		ALLERGY	 ug/mL	Both	2.16	2.73	

Looking at the example units columns, clearly the second term listed with a *Property* of **MCnc** is the correct match. If, however, there were no example units in LOINC, how would you determine the correct *Property*? Based on the parts of the unit of measure ug and mL, we know that the correct LOINC *Property* is a mass-based concentration.

3.2 Qualitative Allergy Terms

For the allergy domain, **PrThr** (presence or threshold) accounts for all but three of the qualitative *Property* terms, with **Ord** being the only *Scale* associated with them. The *Property* **PrThr** indicates whether an analyte is present, or the graded amount of analyte present, as seen with RAST class terms.

[15583-8](#) Vaccinium myrtillus Ab.IgE.RAST class:PrThr:Pt:Ser:Ord:

LCN: Blueberry IgE Ab RAST class [Presence] in Serum

Example answer list: 0, 1, 2, 3, 4, 5, 6

[51546-0](#) Chicken feather Ab.IgG:PrThr:Pt:Ser:Ord:

Chicken feather IgG Ab [Presence] in Serum

Example answer list: Positive, Negative

4 System

The LOINC *System* represents the specimen being tested, which could be derived from the patient or the environment. Matching the *System* for allergy terms is more straightforward than some of the other LOINC domains because the set of values is more limited, and nearly all allergy LOINC terms have one of the three *System* values shown in the table below.

Common allergy Systems	Usage notes
Ser (serum)	The vast majority of LOINC allergy codes have Ser as the <i>System</i> .
Bld (blood)	The <i>System</i> Bld is used for terms that represent leukocyte histamine release testing, which measures the amount of histamine released by leukocytes in peripheral blood in the presence of an allergen.
WBC (white blood cells)	The <i>System</i> WBC is used for terms that represent leukotriene release testing, which measures the amount of leukotriene released by leukocytes in the presence of an allergen.

5 Method

Nearly 90% of LOINC terms in the allergy domain do not specify *Method*. For the remaining 10%, the two primary *Methods* are Multidisk, for terms representing combination allergy tests, and LHR (leukocyte histamine release), for terms representing allergen-triggered histamine release testing. Other methods used for allergy testing include Immune diffusion (ID), Immunoassay (IA), and Immunofluorescence (IF).

6 Allergen Panels

LOINC panels are used for ordering tests for multiple allergens, where each allergen is resulted separately. Note this differs from the combination testing described in [section 2.3](#) above, in which multiple allergens are included in the test but individual allergens are not distinguished as individual results.

Multiplex allergen testing allows for the detection and quantification of multiple allergies in a single assay and can be tailored to a set of allergens appropriate for an individual patient. At present, LOINC provides a single multiplex term, [94593-1](#) Multiplex allergen panel - Serum or Plasma, to use as the order for a multiplex test. This is a “naked panel,” which means it does not contain any child LOINC terms and can be used to order any combination of individual allergens. Individual allergen results would be reported using the appropriate LOINC codes for each allergen tested and resulted.

The vast majority of LOINC allergy panels, other than the multiplex panel, were created prior to 2010 and contain specific sets of required allergens based on the common allergy panels available in the U.S. at the time, which were primarily from one manufacturer. Following are screenshots of a few such panels.

LOINC CODE

48839-5

LONG COMMON NAME

Respiratory allergen panel, US - Inland northwest a - Serum

LOINC STATUS

Active

Panel Hierarchy

Details for each LOINC in Panel

LHC-Forms

LOINC	Name	R/O/C	Cardinality	Example UCUM Units
48839-5	Respiratory allergen panel, US - Inland northwest a - Serum			
6228-1	Red top grass IgE Ab [Units/volume] in Serum	R		k[IU]/L
7155-5	Boxelder IgE Ab [Units/volume] in Serum	R		k[IU]/L
15284-3	Grey Alder IgE Ab [Units/volume] in Serum	R		k[IU]/L
6078-0	Cockroach IgE Ab [Units/volume] in Serum	R		k[IU]/L
51662-5	Core respiratory allergens panel - Serum			
6095-4	American house dust mite IgE Ab [Units/volume] in Serum	R		k[IU]/L
6096-2	European house dust mite IgE Ab [Units/volume] in Serum	R		k[IU]/L
6833-8	Cat dander IgE Ab [Units/volume] in Serum	R		k[IU]/L
6098-8	Dog dander IgE Ab [Units/volume] in Serum	R		k[IU]/L
6020-2	Alternaria alternata IgE Ab [Units/volume] in Serum	R		k[IU]/L
6025-1	Aspergillus fumigatus IgE Ab [Units/volume] in Serum	R		k[IU]/L
6075-6	Cladosporium herbarum IgE Ab [Units/volume] in Serum	R		k[IU]/L
19113-0	IgE [Units/volume] in Serum or Plasma	O		[IU]/L
15283-5	Silver Birch IgE Ab [Units/volume] in Serum	R		k[IU]/L
6189-5	White Oak IgE Ab [Units/volume] in Serum	R		k[IU]/L
6090-5	Cottonwood IgE Ab [Units/volume] in Serum	R		k[IU]/L
6183-8	Mugwort IgE Ab [Units/volume] in Serum	R		k[IU]/L
6234-9	Saltwort IgE Ab [Units/volume] in Serum	R		k[IU]/L
7604-2	Common Pigweed IgE Ab [Units/volume] in Serum	R		k[IU]/L

LOINC CODE

49222-3

LONG COMMON NAME

Food allergen panel - Serum

LOINC STATUS

Active

Panel Hierarchy

Details for each LOINC in Panel

LHC-Forms

LOINC	Name	R/O/C	Cardinality	Example UCUM Units
49222-3	Food allergen panel - Serum			
6106-9	Egg white IgE Ab [Units/volume] in Serum	R		k[IU]/L
6206-7	Peanut IgE Ab [Units/volume] in Serum	R		k[IU]/L
6276-0	Wheat IgE Ab [Units/volume] in Serum	R		k[IU]/L
6273-7	Walnut IgE Ab [Units/volume] in Serum	R		k[IU]/L
6082-2	Codfish IgE Ab [Units/volume] in Serum	R		k[IU]/L
7258-7	Cow milk IgE Ab [Units/volume] in Serum	R		k[IU]/L
6248-9	Soybean IgE Ab [Units/volume] in Serum	R		k[IU]/L
6087-1	Corn IgE Ab [Units/volume] in Serum	R		k[IU]/L
6246-3	Shrimp IgE Ab [Units/volume] in Serum	R		k[IU]/L
7691-9	Scallop IgE Ab [Units/volume] in Serum	R		k[IU]/L
6076-4	Clam IgE Ab [Units/volume] in Serum	R		k[IU]/L

We are reassessing our approach to allergy panels given Because manufacturers' panels may contain different allergen results, specific panels might evolve over time, and there is minimal LOINC representation of non-U.S. allergen panels, LOINC is currently reassessing our approach to allergy panels.

Two options we are considering are:

1. Relaxing the required/optional/conditional values for each result in the panel and expanding the panel definition to explicitly include more LOINC terms or allow for the addition of terms in local panels. For example, we could remove the requirement to include all the foods in the Food allergen panel, indicating results that are optional, and add additional LOINC terms for common allergens such as sesame and almond.
2. Removing the panel children to create "naked panels", which would allow the same panel code to be used for the order across laboratories, even if specific allergens tested vary. With this option, laboratories would simply use the appropriate LOINC codes for each specific allergen in their local panel to report the individual results.

We welcome your input on this topic. Please use our [Contact LOINC](#) page to get in touch. Also, please use this page to request LOINC codes you see are missing for Allergy panels and testing.

Part 3: Mapping Process and Practice Exercises

1 Mapping Validation, Maintenance, and Publication

For optimal success, consideration must be given to how mappings will be validated and maintained over time as both your local testing and the LOINC standard evolve. You will need to develop a plan to make mappings available to the people and systems that need them. Recognize that different use cases for mapping have different requirements. One approach will not satisfy all. In this section, recommendations for addressing these issues are discussed.

1.1 Validating your Mappings

Mapping validation is an important step. Even when highly qualified professionals perform the mapping, errors and inconsistencies are probable.⁵ Be sure to take advantage of mappings provided by reference laboratories and IVD vendors and confirm them for use in your context, keeping in mind that those mappings are informative and not prescriptive.

To increase confidence in the accuracy and consistency of your mappings, multiple experts should be involved. To minimize the potential for bias, each expert could independently map each test, followed by a comparison and reconciliation of discrepancies. Resources may not support this level of mapping redundancy; in which case we still recommend having at least one other reviewer confirm the LOINC choices. This review could be across all mappings, or even a subset of tests with different analytes, methods, and units of measure.

In Mapping projects, it can be beneficial to have two or three levels of mapping, with a different developer for each level. The first group would map the simple, easy to identify tests. The next group would handle the more detailed mapping data. Examples of this would be those tests that need further research, or semantic equivalency in mapping by creating heuristics to explain a mapping choice (specification of cardinality, hierarchical level, and semantic meaning). You can increase the levels of mapping based on the identified complexity in the process, which can vary from one project to another. Finally, one or more QA levels, depending on the project.

Of special note is a Regenstrief Institute service called the Mapping Validity Checker, which is a semi-automated algorithm that checks LOINC mappings for each test in a test catalog. The algorithm looks for inconsistencies and probable errors in the mappings and returns a detailed report highlighting areas in need of further review. Where possible, the report provides suggestions for choosing the correct LOINC code. The detailed feedback and consistency

⁵ Lin MC, Vreeman DJ, McDonald CJ, Huff SM. Correctness of Voluntary LOINC Mapping for Laboratory Tests in Three Large Institutions. AMIA Annu Symp Proc. 2010 Nov 13;2010:447-51. [PMID: [21347018](https://pubmed.ncbi.nlm.nih.gov/21347018/)]

checking represents an exceptional value, even for expert mappers. (Regenstrief has yet to see a mapping file completely free of issues.)

Lastly, for the highest degree of confidence in your mappings, consider working directly with Regenstrief's LOINC staff on a special project basis. Such an arrangement has been used successfully by IVD vendors, professional associations, government agencies, and other organizations. With dedicated resources from Regenstrief, this approach is especially helpful where new LOINC content is likely needed (e.g., new tests) and a high degree of confidence in the mappings is desired.

1.2 Maintaining your Mappings

Mappings of local tests to LOINC terms must be regularly maintained to keep pace with changes in IVD testing, clinical protocols, information systems, and LOINC. There are several types of changes that can impact LOINC mapping, but most fall within one of two categories.

1.2.1 Test Updates

These may be the addition of one or more new tests to an IVD vendor's portfolio, a laboratory's test menu, or an information system (e.g., a health provider contracts with a new laboratory for patient testing needs). Test menu and information system changes may occur when a new reference laboratory is used, or a test previously done onsite is now sent to a reference laboratory. Discontinuation of tests and updates to existing tests may also occur. Updates may occur when a laboratory changes IVD vendors, testing methods, or specimens upon which those tests are done. An IVD vendor may expand not only their assay menu, but also assays on test platforms or how the test is performed on a different type of specimen (e.g., with point of care testing), or how a test is reported. Changes in clinical protocols with new calculations, methods, or tests may warrant changes to the test menu, information system, or LOINC maps. Although one wouldn't expect LOINC code changes if the test system isn't changed, transitions to another information system (LIS, her, etc.) may necessitate different LOINC maps if the test structure, build changes, and/or mapping functionality is different in the new system.

1.2.2 Release Related Changes

LOINC publishes new releases (at least) twice a year. Each LOINC release includes new and previously released LOINC terms, some of which may have been updated since the previous release. Users should review each release for changes in status for all LOINC terms they are using. If a previously active LOINC term you mapped to has now become discouraged or deprecated, you should review the documented rationale for the status change and update the mapping as appropriate. Regenstrief Institute and the LOINC Committee recommend users update to the current version of LOINC within 90 days of its publication. In cases where a terminology provider or other third party is used for mapping, their LOINC updates should occur within 90 days of publication and their customers should update as soon as possible after they are available.

1.2.3 Types of Maintenance Changes to Make

The following are the types of changes LOINC users should consider when assessing whether updates to LOINC maps are needed with a release.

1.2.3.1 New LOINCs

For new LOINC codes, are there more specific LOINC codes that match to your test information? For example, a new method specific LOINC may now be available for your IVD test. If it provides more granular information for downstream users and computer systems, it should be updated.

A new LOINC code may be mapped to a result for which a mapping did not previously exist. As LOINC grows, many such gaps are expected to be filled.

1.2.3.2 Deprecated or Discouraged LOINC codes

These should be reviewed to see if any assays or results have an existing LOINC map with these status changes. Results with these maps will often have two types of changes. One is removal of the map that has changed. The second is often mapping to a new LOINC. Many deprecated LOINC codes include a “map to” recommended new LOINC.

Note: Developers may still include deprecated or discouraged LOINC codes in value sets for querying historic data mapped to these LOINC codes to ensure queries return all possible lab results over a specified time period. It is good practice for users to assess the impact of any LOINC changes for their uses and downstream customer uses as well.

1.2.3.3 Updates to Existing LOINC codes

Updates may occur to LOINC axes, names, status changes, or related information parts of a release. Impacts can vary depending on individual user needs and contexts. (E.g., a typo discovered in a *Component* requires additional updates to LCN and Display Name by the IVD vendor, terminology vendor, or reference laboratory.)

The two primary ways to find newly edited LOINC terms are:

1. Filter on the current version number in the VersionLastChanged field in the LOINC table (available from the [LOINC download page](#)). Once you have filtered on this field, the value in the CHNG_TYPE field indicates whether the term is new, discouraged, deprecated, or updated. (See [Appendix A, LOINC Database Structure](#) in the LOINC Users' Guide for descriptions of CHNG_TYPE values.)
2. Download the [LOINC Table Changes File](#).

For more details regarding term statuses (active, trial, discouraged, and deprecated) as well as other editorial policies, see [Section 11](#), Editorial Policies and Procedures in the LOINC Users' Guide.

KEY POINT

- We strongly advise updating your local system with the latest LOINC version within 90 days of each release.

1.2.3.4 Database Maintenance

From the local database maintenance standpoint, consider the following recommendations:

- Anytime there is a change in the display name, specimen type, reporting format, units of measure, workstation assignment, or interpretive data, there is potential for a LOINC attribute change. Existing mapping should be reviewed when such a change is implemented.
- If instrument downtime or a kit change occurs, the LOINC mappings for the tests run by the substitute bench or referral lab performing the temporary downtime runs should be reviewed. If the LOINC is “stamped” on an open result field and a different LOINC is to be used, the predetermined override procedure (which varies from LIS to LIS) should be invoked.
- If any local tests are mapped to a LOINC term with *System XXX* or no *Method* (i.e., methodless), consider looking for new codes with a more accurate *System* or *Method* with each new LOINC release.

1.3 Mapping Publication

After completing your mapping validation, you will be ready to share them with the people and systems who will put them into action. Consideration should be given in advance to the full lifecycle of these mappings. Different situations will require different processes.

It's important to recognize that you will almost inevitably have to inactivate some mappings and update to superseding codes. Consider where the history logs of those mappings should be stored. (In the source systems that use them? Offline in your mapping environment?) Consider, too, the optimal communication and approval process for publication when the mappings are finalized. You will also want to determine the format of the mapping file, including its structure and contents (e.g., which data fields). RELMA has a highly configurable set of features for exporting mappings. (See the Exporting Local Terms and their LOINC Mappings section in the RELMA Manual.)

For IVD vendors, we strongly recommend publishing the mappings of test codes to LOINC codes for their customers using the newly developed LIVD specification described below.

Next, depending on where your mappings occur and are used in the flow of laboratory data, you may need to alert downstream users of any mapping changes described above. If downstream users are also maintaining their LOINC maps, they may wish to include these changes in their tests.

A single lab result and its LOINC map can have widespread impacts. Updating LOINC codes in all systems helps prevent errors in system interfaces, queries, reports, clinical decision support, and other functionality dependent on LOINC codes. Consider a reference laboratory performing a test on a new IVD instrument or method. They update their existing LOINC code to a different LOINC code reflecting the new method. The change is communicated to their thousands of end users such as smaller laboratories, hospitals, or provider practices. If this test is a public health reportable condition, the laboratory must update all the public health jurisdictions to which they report the LOINC change to avoid errors in electronic laboratory reporting (ELR) or electronic case reporting (eCR) by physicians using their results. If the lab result is part of physician, hospital, or long-term care facility quality measure reporting, queries and reports should be reviewed to ensure the updated LOINC is in the value set being used. If not, these lab results may no longer be included in queries and reports, and penalties may occur as a result of incorrect reporting.

Many clinical decision support tools leverage LOINC codes for calculations, patient care, financial, operational, and other healthcare-related decisions. A *Method* change that produces values 5% different than previous results can have clinically significant impacts, not only to end-user decisions and patient care, but also computerized calculations, alerts, and recommendations. Harmful or fatal patient care decisions could result. Maintenance and updates to mappings should occur on a regular basis.

1.3.1 LIVD Specification for Vendor to LOINC Mappings

LIVD (LOINC In Vitro Diagnostic) specifies a standard digital format for IVD vendors to publish the LOINC terms associated with each of their test results. The LIVD specification is published by the IVD Industry Connectivity Consortium (IICC). It was created as a resource for publishing and exchanging mappings of LOINC terms for IVD test results. IVD vendors are well positioned to identify the appropriate LOINC terms for their test results because they possess all the relevant information about the test. When paired with expert LOINC knowledge and review, IVD vendors can create high quality mappings that can then be used by their customers to achieve more consistent coding choices across the industry. As IVD vendors make these LIVD format mappings available, the reduced effort and expense of choosing the correct LOINC term will dramatically boost the ability of labs to implement standard terminology.

For more information regarding the LIVD project, see the LIVD page on the IICC website.

KEY POINT

- If you are an IVD vendor mapping your tests to LOINC, we encourage publishing your mappings using the LIVD format.

2 Putting it All Together – Mapping Practice

2.1 Quiz

Time to put the knowledge gained to the test. Below are exercises to help further illustrate various practical aspects of mapping local terms to LOINC codes. An answer key is provided at the end of this guide.

1. Identify the missing pieces necessary for an accurate mapping.

Local test: Leukocyte concentration (10⁹/L)

- ☐ *Analyte (Component)*
- ☐ *Property*
- ☐ *Time Aspect*
- ☐ *Specimen type (System)*
- ☐ *Units of measure*
- ☐ *Method*

2. Which LOINC best matches the local test? (Answer options follow this model:

LOINC Code: *Component:Property:Timing:System:Scale:Method*)

Local test: HIV-2 Antibody - Serum - Rapid Immunoassay - Present/Absent

- ☐ 31073-0: HIV 2 Ab band pattern:Imp:Pt:Ser:Nom:IB
- ☐ 51786-2: HIV 2 Ab Signal/Cutoff:RelACnc:Pt:Ser/Plas:Qn:IA
- ☐ 33807-9: HIV 2 Ab.IgG:PrThr:Pt:Ser:Ord:
- ☐ 81641-3: HIV 2 Ab:PrThr:Pt:Ser/Plas/Bld:Ord:IA.rapid
- ☐ 5224-1: HIV 2 Ab:ACnc:Pt:Ser/Plas:Ord:IA

3. Why is the LOINC below a poor choice for the local test?

LOINC: 2839-9: Progesterone:MCnc:Pt:Ser/Plas:Qn:

Local test: Progesterone Concentration - Urine (24H)

- ☐ *Analytes* are different
- ☐ *Properties* do not align
- ☐ *System* mismatch
- ☐ *Timing* difference

4. What further information is necessary before attempting to map the local term below?

Local test: Vanillylmandelic Acid, 24 Hr Urine

- ☐ *Analyte (Component)*
- ☐ *Property*
- ☐ *Time Aspect*
- ☐ *Specimen type (System)*
- ☐ *Units of measure*
- ☐ *Method*

5. Which LOINC best matches the local test? (Answer options follow this model: LOINC Code: *Component:Property:Timing:System:Scale:Method*)

Local test: Influenza Virus B Ab, IgM concentration - Serum - ELISA

- ☐ 17016-7: Influenza virus B Ab.IgM:ACnc:Pt:Ser:Qn:
- ☐ 86570-9: Influenza virus B Ab.IgM:PrThr:Pt:Ser:Ord:IA
- ☐ 57328-7: Influenza virus B Ab.IgM^1st specimen:ACnc:Pt:Ser:Qn:IA
- ☐ 47315-7: Influenza virus B Ab.IgM:ACnc:Pt:Ser:Qn:IA
- ☐ 49078-9: Influenza virus B Ab.IgM:MCnc:Pt:CSF:Qn:IA

6. Why is the LOINC below a poor choice for the local test?

LOINC: 82062-1 - Beef+Chicken meat+Pork+Turkey meat Ab.IgE RAST
class:PrThr:Pt:Ser:

Local test: Beef+Poultry+Pork Allergy in Serum (Presence)

- ☐ *Analytes are different*
- ☐ *Properties do not align*
- ☐ *System mismatch*
- ☐ *Method difference*

7. Identify the missing pieces preventing an accurate mapping.

Presence of Bacilliform bacteria

- ☐ *Analyte (Component)*
- ☐ *Property*
- ☐ *Time Aspect*
- ☐ *Specimen type (System)*
- ☐ *Units of measure*
- ☐ *Method*

8. Which of the following LOINC codes should a vendor provide for their assay? (Answer options follow this model: LOINC Code: *Component:Property:Timing:System:Scale:Method*)

IIVD Vendor assay: Qualitative screen 11-Hydroxy delta-9 tetrahydrocannabinol.

This assay can be set up to report on either Serum or Urine. It is expected that laboratory customers would either build a single test result for the specific specimen tested or a single test result to accommodate values from either specimen if testing may be performed on both.

- ☐ 78752-3: 11-Hydroxy delta-9 tetrahydrocannabinol:PrThr:Pt:Body fld:Ord:Screen
- ☐ 80111-8: 11-Hydroxy delta-9 tetrahydrocannabinol:MCnc:Pt:Bld:Qn:Confirm
- ☐ 70178-9: 11-Hydroxy delta-9 tetrahydrocannabinol:PrThr:Pt:Bld:Ord:
- ☐ 40864-1: 11-Hydroxy delta-9 tetrahydrocannabinol:MCnc:Pt:Ser/Plas:Qn:
- ☐ 80112-6: 11-Hydroxy delta-9 tetrahydrocannabinol:MCnc:Body fld:Qn:Confirm
- ☐ 78753-1: 11-Hydroxy delta-9 tetrahydrocannabinol:PrThr:Pt:Ser/Plas:Ord:Screen
- ☐ 48934-4: 11-Hydroxy delta-9 tetrahydrocannabinol:MCnc:Pt:XXX:Qn:
- ☐ 78754-9: 11-Hydroxy delta-9 tetrahydrocannabinol:PrThr:Pt:Urine:Ord:Screen
- ☐ 48935-1: 11-Hydroxy delta-9 tetrahydrocannabinol:PrThr:Pt:XXX:Ord:
- ☐ 78751-5: 11-Hydroxy delta-9 tetrahydrocannabinol:MCnt:Pt:Hair:Qn:

9. Match the *Property* to the unit

<i>Property</i>		<i>Answer</i>	<i>Unit</i>	
A	MCnc	<i>Example: A-3</i>	1	log copies/mL
B	NFr		2	mg/d
C	SCnt		3	mg/dL
D	Naric		4	umol/mol{creatinine}
E	LnCnc		5	#/area
F	MRat		6	nmol/g
G	SRto		7	%

10. Which *Component* best represents a single test that differentiates between two or more analytes?

- ☐ HIV 1+2 Ab+HIV1 p24 Ag
- ☐ Dengue virus 1 & 2 & 3 & 4 RNA
- ☐ Foot and mouth disease virus serotype
- ☐ Dengue virus 1+2+3+4 & Zika virus Ab.IgA+IgG+IgM
- ☐ HIV 1 & 2 Ab
- ☐ 11-Dehydro thromboxane beta 2/Creatinine

2.2 Answer Key

1. Identify the missing pieces necessary for an accurate mapping.

Local test: Leukocyte concentration (10⁹/L)

- ☐ *Analyte (Component)* – Leukocyte is sufficient, but it is possible this is for a more specific target, e.g., Leukocytes.disintegrated or Leukocytes^{^^}corrected for nucleated erythrocytes
- ☐ *Property* – based on the units provided (10⁹/L), we know this is Number Concentration (NCnc)
- ☐ *Time Aspect* – depending on the (missing) specimen, this could be point in time (Pt) for blood or 24 hour (24H) for urine
- ✓ **Specimen type (System)** – **System** is a required element of every LOINC, and can distinguish similar terms
- ☐ Units of measure
- ✓ **Method** – while not required, if the **Method** is known (and relevant), this could narrow the list of potential matches

See [Section 2.2](#) Minimum Test Information Necessary to Map.

2. Which LOINC best matches the local test?

Local test: HIV-2 Antibody - Serum - Rapid Immunoassay - Present/Absent

- ☐ 31073-0: HIV 2 Ab band pattern:Imp:Pt:Ser:Nom:IB
- ☐ 51786-2: HIV 2 Ab Signal/Cutoff:RelACnc:Pt:Ser/Plas:Qn:IA
- ☐ 33807-9: HIV 2 Ab.IgG:PrThr:Pt:Ser:Ord:
- ✓ **81641-3: HIV 2 Ab:PrThr:Pt:Ser/Plas/Bld:Ord:IA.rapid**
- ☐ 5224-1: HIV 2 Ab:ACnc:Pt:Ser/Plas:Ord:IA

81641-3 is the only term listed that matches the **Component** (HIV 2 Ab), the **Property** (PrThr, indicating Presence or Threshold), and I (IA. rapid for Rapid Immunoassay).

See [Section 2](#) Choosing the Right LOINC Term (Mapping).

3. Why is the LOINC below a poor choice for the local test?

LOINC: 2839-9: Progesterone:MCnc:Pt:Ser/Plas:Qn:

Local test: Progesterone Concentration - Urine (24H)

- ☐ *Analytes* are different
- ☐ *Properties* do not align
- ✓ **System mismatch (Serum/Plasma vs. Urine)**
- ✓ **Timing difference (Point in time vs. 24-hour collection)**

See [Section 2.1](#) General mapping principles.

4. What further information is necessary before attempting to map the local test below?

Local test: Vanillylmandelic Acid, 24 Hr Urine

- ☐ *Analyte (Component)*
- ✓ **Property** – As shown, the local test provides no indication as to whether this is a concentration, rate, or other type of measure. **Property** is a required element and must be known to find an accurate match.
- ☐ *Time Aspect*
- ☐ Specimen type (*System*)
- ✓ **Units of measure** – not critical, but in the absence of **Property**, could help narrow the list of potential matches
- ✓ **Method** – while also not critical, if the **Method** is known (and relevant), this could narrow the list of potential matches.

See [Section 2.2](#) Minimum Test Information Necessary to Map.

5. Which LOINC best matches the local test?

Local test: Influenza Virus B Ab.IgM concentration - Serum - ELISA

- ☐ 17016-7: Influenza virus B Ab.IgM:ACnc:Pt:Ser:Qn:
- ☐ 86570-9: Influenza virus B Ab.IgM:PrThr:Pt:Ser:Ord:IA
- ☐ 57328-7: Influenza virus B Ab.IgM^{1st specimen}:ACnc:Pt:Ser:Qn:IA
- ✓ **47315-7: Influenza virus B Ab.IgM:ACnc:Pt:Ser:Qn:IA**
- ☐ 49078-9: Influenza virus B Ab.IgM:MCnc:Pt:CSF:Qn:IA

47315-7 is the only LOINC listed that matches the *Component* (Influenza virus B Ab.IgM), matches the Serum specimen, has a “concentration” *Property* (ACnc), and specifies an Immunoassay *Method*.

See [Section 2.1](#) General mapping principles.

6. Why is the LOINC below a poor choice for the local test?

LOINC: 82062-1 - Beef+Chicken meat+Pork+Turkey meat Ab.IgE RAST
class:PrThr:Pt:Ser:

Local test: Beef+Poultry+Pork Allergy in Serum (Presence)

- ✓ ***Analytes are different (Chicken and Turkey meat vs. Poultry)***
- ☐ *Properties* do not align
- ☐ *System* mismatch
- ☐ *Method* difference

See [Section 2.4.1](#) *Component*.

7. Identify the missing pieces preventing an accurate mapping.

Presence of Bacilliform bacteria

- ☐ *Analyte (Component)* – “Bacilliform bacteria” clearly indicates the target
- ☐ *Property* – “Presence” provides the information necessary to know this is PrThr
- ☐ *Time Aspect* – Knowing this test detects bacteria, one could deduce the LOINC axis is Point in Time. Although nice to have, time is not required as part of the minimum data necessary to accurately map in this case.
- ✓ ***Specimen type (System)* – *System* is a required element of every LOINC, and can distinguish similar terms, which differ by specimen type. However, many microbiology tests may be designed by the IVD vendor or in the LIS to accommodate many different specimen sources or specimen types. In these cases, a specimenless LOINC “XXX” may be mapped. Either way, this information is needed for accurate mapping.**
- ☐ Units of measure
- ✓ ***Method* – if the *Method* is known (and relevant), this could narrow the list of potential matches. Additionally knowing if this test is performed as part of urinalysis versus microbiology, would also help narrow down the choices as several options are performed on urine sediment with automated methods. IVD vendors which have instrumentation where this may be reported should include these LOINC codes in their mappings.**

See [Section 2.2](#) Minimum Test Information Necessary to Map.

8. Which of the following LOINC codes should a vendor provide for their assay?

IVD Vendor assay: Qualitative screen 11-Hydroxy delta-9 tetrahydrocannabinol. This assay can be set up to report on either Serum or Urine. It is expected that laboratory customers would either build a single test result for the specific specimen tested or a single test result to accommodate values from either specimen if testing may be performed on both.

- ☐ 78752-3: 11-Hydroxy delta-9 tetrahydrocannabinol:PrThr:Pt:Body fld:Ord:Screen
- ☐ 80111-8: 11-Hydroxy delta-9 tetrahydrocannabinol:MCnc:Pt:Bld:Qn:Confirm
- ☐ 70178-9: 11-Hydroxy delta-9 tetrahydrocannabinol:PrThr:Pt:Bld:Ord:
- ☐ 40864-1: 11-Hydroxy delta-9 tetrahydrocannabinol:MCnc:Pt:Ser/Plas:Qn:
- ☐ 80112-6: 11-Hydroxy delta-9 tetrahydrocannabinol:MCnc:Body fld:Qn:Confirm
- ✓ **78753-1: 11-Hydroxy delta-9 tetrahydrocannabinol:PrThr:Pt:Ser/Plas:Ord:Screen**
- ☐ 48934-4: 11-Hydroxy delta-9 tetrahydrocannabinol:MCnc:Pt:XXX:Qn:
- ✓ **78754-9: 11-Hydroxy delta-9 tetrahydrocannabinol:PrThr:Pt:Urine:Ord:Screen**
- ☐ 48935-1: 11-Hydroxy delta-9 tetrahydrocannabinol:PrThr:Pt:XXX:Ord:
- ☐ 78751-5: 11-Hydroxy delta-9 tetrahydrocannabinol:MCnt:Pt:Hair:Qn:

78753-1 and 78754-9 are the only two terms listed that match the *Method* (screen) and *Systems* (Serum or Urine) of the two ways the IVD assay may be

resulted. While the IVD vendor will provide both LOINC codes, it is assumed only one of the specimens is used for testing by the laboratory. In this approach, the LIS build will have a test result for the Serum result and value or the Urine result and value, and the laboratory professional will need to choose which of the two vendor maps matches their specimen analyzed.

Alternatively, since this is a qualitative test result and the vendor is approved to perform testing on either specimen with the assay and results are reported in the same manner for both (i.e. there is no capability to set up the assay differently on the instrument for the different specimen types), there may be cases where 48935-1: 11-Hydroxy delta-9 tetrahydrocannabinol:PrThr:Pt:XXX:Ord: also needs to be provided by the IVD vendor. The LIS build may have a single result for Qualitative screen 11-Hydroxy delta-9 tetrahydrocannabinol and its qualitative value and the specimen may be indicated as a separate result in the test build. In this regard, given the multiple different specimens, a specimenless LOINC is needed. No specimenless screening LOINC exists, so which LOINC should be mapped? 48935-1 is the only LOINC that is qualitative that addresses results on multiple specimens. This approach is more common with qualitative multiple specimen results, as quantitative results that can be performed on multiple specimens frequently will require different result builds, one for each reference range as they often differ.

This can get a bit tricky as an IVD vendor may not know if one or two different specimens are involved in the setup for their assay at each laboratory and thus which LOINC may be needed for a particular laboratory. By providing all three LOINC codes in this example, the different LOINC codes corresponding to the different configurations needed by their customers are all available for the customer to choose which best corresponds to their test setup.

See [Section 1.1.1](#) In Vitro Diagnostics (IVD) Vendors.

9. Match the *Property* to the unit

<i>Property</i>		<i>Answer</i>	<i>Unit</i>	
A	MCnc	A-3	1	log copies/mL
B	NFr	B-7	2	mg/d
C	SCnt	C-6	3	mg/dL
D	Naric	D-5	4	umol/mol{creatinine}
E	LnCnc	E-1	5	#/area
F	MRat	F-2	6	nmol/g
G	SRto	G-4	7	%

See [Section 2.4.3](#) *Scale and Property*.

10. Which *Components* best represent a single test differentiation between two or more analytes?

- ☐ HIV 1+2 Ab+HIV1 p24 Ag
- ✓ **Dengue virus 1 & 2 & 3 & 4 RNA**
- ☐ Foot and mouth disease virus serotype
- ✓ **Dengue virus 1+2+3+4 & Zika virus Ab.IgA+IgG+IgM**
- ✓ **HIV 1 & 2 Ab**
- ☐ 11-Dehydro thromboxane beta 2/Creatinine

Remember that analytes separated by ‘&’ CAN be differentiated, while those with ‘+’ cannot. As a result, HIV 1+2Ab+HIV1p24Ag is an assay where one, two, or all three may be reactive by an assay, but it doesn’t tell you which one or ones are reacting. The foot and mouth disease response indicates a single serotype. The last response is a ratio, which does involve two analytes, but with a single calculated value. The Dengue virus 1 & 2 & 3 & 4 RNA differentiates which of the 4 types are present, as does the HIV 1 & Ab. The Dengue and Zika test distinguish each of these analytes as well. Although within the Zika assay the individual antibodies are not differentiated, and there is also no differentiation within the Dengue virus assay.

See [Section 2.4.2.3](#) *Ampersand vs. Plus*.

3 Resources

3.1 LOINC website

The loinc.org website is the home to everything LOINC. It features numerous resources for both new and experienced users. In addition to the more prominent tools mentioned here, the LOINC website also has a [get started](#) primer for new users and educational [videos](#) for various skill levels.

For many of the following resources, a free LOINC login is required. You can sign up at loinc.org/join.

3.1.1 Download LOINC

You can download the entire LOINC database from loinc.org/downloads. This single zip archive contains all the necessary table files, plus a number of ancillary files which may prove useful in your particular use of LOINC.

3.1.2 SearchLOINC

SearchLOINC (loinc.org/search) is a handy, web-based application that allows you to locate LOINC concepts using a robust system of filters, synonymy, and field syntax. Search results can be exported in a number of formats.

3.1.3 LOINC Hierarchy Browser

You can view several of the hierarchical structures available in LOINC with this tool, available at loinc.org/tree. An ideal complement to SearchLOINC, you can see how LOINC concepts are arranged by *Component*, *Class*, *System*, and *Method*. There are also special hierarchies for Panels and the LOINC Document Ontology.

3.1.4 LOINC Terminology Service using HL7 FHIR

LOINC content can be retrieved programmatically with this service using the FHIR standard for data resources. This API supports numerous versions of LOINC and translations. Complete documentation is available at loinc.org/fhir.

3.2 Documentation

3.2.1 LOINC Knowledge Base

This repository collects all LOINC documentation in a single location at loinc.org/kb. You can easily search to find what you need. In addition to online versions of the following documentation, the knowledge base has LOINC Release Notes, Frequently Asked Questions, common abbreviations used in LOINC, and other helpful information.

3.2.2 LOINC Users' Guide

The LOINC Users' Guide is the definitive document about LOINC. It explains the structure of the table, its rationale, and the rules LOINC uses for naming test results. This PDF is included in the Complete LOINC file found at loinc.org/downloads.

3.2.3 RELMA Users' Manual

If you need help with Regenstrief's LOINC Mapping Assistant, this PDF is your source for answers. It can be downloaded via loinc.org/relma.

3.2.4 LOINC Terminology Service Documentation

We have extensive information on this API at loinc.org/fhir, complete with all available functionality and example requests.

3.2.5 Journal publications

LOINC is a popular topic within biomedical informatics literature. You can find relevant articles in the U.S. National Library of Medicine's PubMed database. The LOINC website also has a curated list of recent [journal articles](#). The following are popular citations.

- Abhyankar S, Demner-Fushman D, McDonald CJ. Standardizing clinical laboratory data for secondary use. *Journal of biomedical informatics*. 2012;45(4):642-650. doi:10.1016/j.jbi.2012.04.012. [PMID: [22561944](#)]
- Vreeman DJ, Hook J, Dixon BE. Learning from the crowd while mapping to LOINC. *J Am Med Inform Assoc*. 2015;22(6):1205-11. doi: 10.1093/jamia/ocv098. [PMID: [26224334](#)]
- Kim H, El-Kareh R, Goel A, Vineet F, Chapman WW. An Approach to Improve LOINC Mapping through Augmentation of Local Test Names. *Journal of biomedical informatics*. 2012;45(4):651-657. doi:10.1016/j.jbi.2011.12.004. [PMID: [22210167](#)]

3.3 Mapping

[RELMA](#) is the Regenstrief LOINC Mapping Assistant, a Windows application to assist with the association of your local test catalog to LOINC codes.

3.4 Requesting New LOINCs or Edits to Existing LOINCs

LOINC codes are primarily created based on community (user) requests. If you cannot find an appropriate LOINC term for your test, check the [LOINC pre-release web page](#) and the terms currently [in queue](#). If you do not see the concept you need, you are welcome to submit a request for a new LOINC term by following the instructions on our [Submissions](#) page.

Likewise, we rely on the LOINC community to help us keep up with the latest scientific knowledge. If you are aware of any changes needed to existing terms, such as *Component* updates, addition of synonyms, or updating term or Part descriptions, you can submit these via our [Contact LOINC](#) form.

3.5 Related standards

3.5.1 UCUM

The Unified Code for Units of Measure (UCUM) is a code system intended to include all units of measures being contemporarily used in international science, engineering, and business. The purpose is to facilitate unambiguous electronic communication of quantities together with their units. <https://ucum.org>

3.5.2 HL7 v2

http://www.hl7.org/implement/standards/product_brief.cfm?product_id=185

3.5.3 HL7 CDA

http://www.hl7.org/implement/standards/product_brief.cfm?product_id=7

3.5.4 HL7 FHIR

http://www.hl7.org/implement/standards/product_brief.cfm?product_id=449

Allergy mapping example questions

Test Number	Test Name	Component Number	Component Mnemonic	Component Name	Pre LOINC Mapping	Post LOINC Mapping
<i>Example</i>	<i>Allergen, Food, Grape</i>	<i>55015</i>	<i>Grape</i>	<i>Allergen, Food, Grape IgE</i>	<i>6841-1 Grape IgE Ab [Units/volume] in Serum</i>	<i>6841-1G rape IgE Ab [Units/volume] in Serum</i>
0055381	Allergen, Food, Egg Whole	55381	Egg Whole	Allergen, Food, Egg Whole IgE	7291-8DWhole Egg IgE Ab [Units/volume] in Serum	7291-8WD hole Egg IgE Ab [Units/volume] in Serum
55151	Allergen, Grass, Bahia	55151	Bahia Grass	Allergen, Grass, Bahia IgE	6034-3DBahia grass IgE Ab [Units/volume] in Serum	6034-3BDahia grass IgE Ab [Units/volume] in Serum
55368	Allergens, Inhalant Multiallergen IgE, Qualita	55368	Inhal Mult Qual	Allergen, Inhalant, Multiallergen IgE	37989-1IDnhalant Allergen Mix 8 (Mugwort+Common silver birch+Cat dander+C herbarum+Dermatophagoides pteronyssinus+Dog dander+Timothy grass+Rye) IgE Ab [Presence] in Serum by Multidisk	37989-1IDnhalant Allergen Mix 8 (Mugwort+Common silver birch+Cat dander+C herbarum+Dermatophagoides pteronyssinus+Dog dander+Timothy grass+Rye) IgE Ab [Presence] in Serum by Multidisk
98641	Allergen, Grass, Cultivated Corn Pollen	98641	Corn Pollen	Allergen, Grass, Cult. Corn Pollen IgE	7248-8DCultivated Corn IgE Ab [Units/volume] in Serum	7248-8CDultivated Corn IgE Ab [Units/volume] in Serum
55078	Allergen, Food, Corn	55078	Corn	Allergen, Food, Corn IgE	D6087-1DCorn IgE Ab [Units/volume] in Serum	6087-1CDorn IgE Ab [Units/volume] in Serum
97658	Allergen, Food, Corn IgG	97658	Corn IgG	Allergen, Food, Corn IgG	6723-1DCorn IgG Ab [Units/volume] in Serum	35534-7CDorn IgG Ab [Mass/volume] in Serum

55037	Allergen, Epidermal s and Animal Proteins, F	55037	Feather Mix	Allergen, Animal, Feather Mix IgE	31161-3DEpidermal Allergen Mix 73 (Chicken feather+Duck feather+Goose feather+Parrot feather) IgE Ab [Units/volume] in Serum by Multidisk	72279-3EDpidermal Allergen Mix 71 (Chicken feather+Duck feather+Goose feather+Turkey feather) IgE Ab [Units/volume] in Serum by Multidisk
50147	Allergen, Fungi and Molds, Trichophyton me	50147	Trichophyton men	Allergen, Fungi/Mold, Tricho. men. IgE	25801-2DTrichophyton mentagrophytes var interdigitale IgE Ab [Units/volume] in Serum	25801-2DTrichophyton mentagrophytes var interdigitale IgE Ab [Units/volume] in Serum
98617	Allergen, Food, Hazelnut (Filbert)	98617	Hazelnut	Allergen, Food, Hazelnut IgE	D6136-6DHazelnut IgE Ab [Units/volume] in Serum	6136-6HDazelnut IgE Ab [Units/volume] in Serum
93506	Allergen, Tree, Hazelnut IgE	93507	DHazel Nut Tree	Allergen, Tree, Hazelnut IgE	D6137-4DHazelnut Pollen IgE Ab [Units/volume] in Serum	6136-6HDazelnut IgE Ab [Units/volume] in Serum
2007211	Allergen, Food, Peanut Components IgE	55024	Peanut	Allergen, Food, Peanut IgE	6206-7DPeanut IgE Ab [Units/volume] in Serum	6206-7PDeanut IgE Ab [Units/volume] in Serum
2007211	Allergen, Food, Peanut Components IgE	2010810	Peanut Ara h 1	Allergen, Food, Severe Peanut Ara h 1	58779-0DPeanut recombinant () 1 IgE Ab [Units/volume] in Serum	58779-0PDeanut recombinant () 1 IgE Ab [Units/volume] in Serum
2007211	Allergen, Food, Peanut Components IgE	2010811	Peanut Ara h 2	Allergen, Food, Severe Peanut Ara h 2	D58778-2DPeanut recombinant (rAra h) 2 IgE Ab [Units/volume] in Serum	D58778-2PDeanut recombinant (rAra h) 2 IgE Ab [Units/volume] in Serum

2007211	Allergen, Food, Peanut Components IgE	2010812	Peanut Ara h 3	Allergen, Food, Severe Peanut Ara h 3	58777-4DPeanut recombinant (rAra h) 3 IgE Ab [Units/volume] in Serum	58777-4PDeanut recombinant (rAra h) 3 IgE Ab [Units/volume] in Serum
2007211	Allergen, Food, Peanut Components IgE	2010813	Peanut Ara h 9	Allergen, Food, Severe Peanut Ara h 9	64965-7DPeanut recombinant (rAra h) 9 IgE Ab [Units/volume] in Serum	64965-7PDeanut recombinant (rAra h) 9 IgE Ab [Units/volume] in Serum
2007211	Allergen, Food, Peanut Components IgE	2010814	Peanut Ara h 8	Allergen, Food, Mild Peanut Ara h 8	D63477-4DPeanut recombinant (rAra h) 8 IgE Ab [Units/volume] in Serum	D63477-4PDeanut recombinant (rAra h) 8 IgE Ab [Units/volume] in Serum
2007211	Allergen, Food, Peanut Components IgE	3002252	Peanut Ara h 6	Allergen, Food, Severe Peanut Ara h 6	90880-6DPeanut recombinant (rAra h) 6 IgE Ab [Units/volume] in Serum	90880-6PDeanut recombinant (rAra h) 6 IgE Ab [Units/volume] in Serum

Answers to examples

Test Number	ARUP Current State	Reason for mismatch	Answers/Comments
<i>Example</i>			
0055381	7291-8		Allergen codes are included as related names linked to the appropriate LOINC Components.
55151	6034-3		To reduce ambiguity, LOINC uses the Latin name of the species of the biological substance that causes the allergy in the Component part of the fully specified name. The common name for the substance is used in the LOINC Long Common Name, Display Name, and other term names. In addition, most LOINC Parts have associated synonyms that are assembled into a list called "Related Names".
55368	37989-1		To clearly represent combinations, the set of allergens tested are enclosed in parenthesis in the LOINC Component with the allergens separated by a '+'. .
98641	7248-8		Some plants have more than one type of allergen, for example, IgE antibodies can develop towards tree pollen and the fruit of the same tree.
55078	6087-1		Some plants have more than one type of allergen, for example, IgE antibodies can develop towards tree pollen and the fruit of the same tree.
97658	35534-7		Not in the guide - IgE vs IgG
55037	31161-3	incorrect component	To clearly represent combinations, the set of allergens tested are enclosed in parenthesis in the LOINC Component with the allergens separated by a '+'. .
50147	25801-2		Not in the guide - specificity of choosing the correct variant of an organism
98617	6136-6		Some plants have more than one type of allergen, for example, IgE antibodies can develop towards tree pollen and the fruit of the same tree.
93506	6136-6	incorrect component	Some plants have more than one type of allergen, for example, IgE antibodies can develop towards tree pollen and the fruit of the same tree.
2007211	6206-7		The <i>Long Common Name</i> will use the common name of that entity, e.g., Peanut, if one exists. In the case of very well specified allergens, the <i>Component</i> will also indicate whether the allergen has a natural source or has been generated via recombinant method by adding the word, native or recombinant , respectively.

2007211	58779-0		The <i>Long Common Name</i> will use the common name of that entity, e.g., Peanut, if one exists. In the case of very well specified allergens, the <i>Component</i> will also indicate whether the allergen has a natural source or has been generated via recombinant method by adding the word, native or recombinant , respectively.
2007211	58778-2		We insert standard acronym names used by most allergen manufacturers as synonyms. These begin with lower case "n" or "r" to distinguish native from recombinant allergens followed by the first 3 letters of the genus (in Latin), a space and the first letter of the species (in Latin).
2007211	58777-4		We insert standard acronym names used by most allergen manufacturers as synonyms. These begin with lower case "n" or "r" to distinguish native from recombinant allergens followed by the first 3 letters of the genus (in Latin), a space and the first letter of the species (in Latin).
2007211	64965-7		We insert standard acronym names used by most allergen manufacturers as synonyms. These begin with lower case "n" or "r" to distinguish native from recombinant allergens followed by the first 3 letters of the genus (in Latin), a space and the first letter of the species (in Latin).
2007211	63477-4		We insert standard acronym names used by most allergen manufacturers as synonyms. These begin with lower case "n" or "r" to distinguish native from recombinant allergens followed by the first 3 letters of the genus (in Latin), a space and the first letter of the species (in Latin).
2007211	90880-6		We insert standard acronym names used by most allergen manufacturers as synonyms. These begin with lower case "n" or "r" to distinguish native from recombinant allergens followed by the first 3 letters of the genus (in Latin), a space and the first letter of the species (in Latin).